

3DGEER: PROJECTIVE-EXACT AND EFFICIENT 3D GAUSSIAN RENDERING FOR GENERIC CAMERAS

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Anonymous submission

ABSTRACT

3D Gaussian Splatting (3DGS) (Kerbl et al., 2023) achieves an influential quality–efficiency trade-off by replacing the projective behavior of 3D Gaussians with local 2D image-plane approximations. This approximation is often acceptable for narrow field-of-view (FoV) imagery, but it becomes a major source of error for generic wide-FoV and fisheye cameras, where projection is strongly nonlinear. Existing extensions either preserve the same projective approximation or obtain projective correctness through ray-based procedures whose association cost limits real-time use. We present 3DGEER, a 3D Gaussian rendering framework that is projective-exact at the per-Gaussian ray-integral level while retaining real-time efficiency for perspective and fisheye camera models. From first principles, we derive a closed-form integral of Gaussian density along a ray, yielding exact per-particle forward transmittance and differentiable optimization under arbitrary camera projections within the standard Gaussian alpha-compositing pipeline. To make this formulation practical, we introduce the *Particle Bounding Frustum (PBF)* for tight frustum-level ray–Gaussian association without BVH traversal, together with the *Bipolar Equiangular Projection (BEAP)*, which supplies a unified angular parameterization, accelerates association, and provides more balanced supervision. Across pinhole and fisheye datasets, 3DGEER improves all reported quality metrics over prior methods, renders $5\times$ faster than existing projective-exact ray-based baselines, and preserves reconstruction quality when evaluated at wider FoVs than those observed during training, establishing a strong real-time rendering baseline for the evaluated generic-camera settings.

3D Gaussian Splatting (3DGS) (Kerbl et al., 2023) has made volumetric particle rendering a central paradigm for neural scene representation because splatting-based methods (Yu et al., 2024a, Yan et al., 2024; Lu et al., 2024; Kheradmand et al., 2024; Charatan et al., 2024; Mallick et al., 2024) combine high visual fidelity with interactive rendering speed. This efficiency originates in EWA splatting (Zwicker et al., 2001), which locally linearizes the camera projection and represents each 3D Gaussian by a 2D Gaussian on the image plane. The approximation is usually tolerable for narrow field-of-view (FoV) views, but it becomes a dominant error source when generic wide-FoV cameras induce strong nonlinear distortion: the true projection is no longer well described by a symmetric 2D Gaussian, particularly in fisheye imagery used in robotics and autonomous-driving perception (Rashed et al., 2021; Courbon et al., 2007; Kumar et al., 2023; Zhang et al., 2016; Yoganani et al., 2019, 2024; Sekkat et al., 2022). Recent differentiable renderers for specific fisheye models (Liao et al., 2024) and more general camera models (Huang et al., 2024b) enlarge the supported camera family, but their rendering and association stages still depend on 2D projective-space approximations rather than a closed-form ray treatment of 3D Gaussian density.

A natural response to projective approximation error is to allocate more particles. Celarek et al. (2025) show that scaling 3DGS to as many as 1M Gaussians can reduce artifacts, thereby questioning whether exact rendering is necessary in some regimes. That evidence, however, is confined to small object-centric scenes. In larger reconstructions with wide-FoV cameras, Gaussian density must adapt across objects, depths, and highly distorted view frustums, so brute-force scaling quickly becomes costly. Our experiments further show that the projective error induced by generic distorted cameras is substantially larger than the error observed in standard narrow-FoV settings: increasing the Gaussian count alone does not close the quality gap in the evaluated fisheye scenes (Fig. 5).

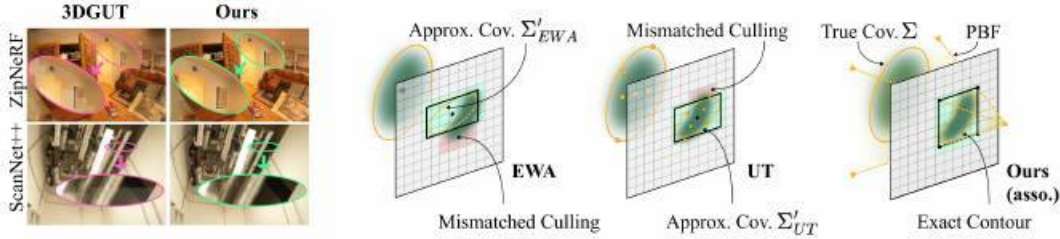


Figure 1: **Linear Approximation Error in Ray-Particle Association.** (Left) UT-based association can produce visible grid-line artifacts when its projective approximation misaligns tiles and particles. (Right) 3DGEER avoids the intermediate conic approximation and computes the bounding structure directly from the true 3D covariance.

Ray-tracing and ray-marching formulations provide the most direct route to projectively correct volumetric particle rendering (Yu et al., 2024b; Condor et al., 2025; Mai et al., 2025; Moënné-Loccoz et al., 2024; Talegaonkar et al., 2025; Gu et al., 2025). Because rays are evaluated in the camera model itself, these approaches naturally accommodate large-FoV projections and avoid image-plane linearization. Their bottleneck is computational rather than geometric: efficient rendering requires associating each ray, or ray bundle, with the relevant 3D particles. BVH-based systems such as EVER (Mai et al., 2025) and 3DGRT (Moënné-Loccoz et al., 2024) accelerate this search, but traversal overhead and limited parallelism prevent pure ray-based renderers from matching the throughput of 3DGS. Hybrid designs such as 3DGUT (Wu et al., 2025) use rasterization to accelerate association for ray-tracing rendering, but the association stage reintroduces projective approximation and can produce grid-line artifacts (Fig. 1 Left). The central technical challenge is therefore to preserve the projective correctness of ray evaluation while recovering the practical association efficiency that makes splatting attractive.

We address this challenge with 3DGEER, a 3D Gaussian Exact and Efficient Rendering method that is *exact in projective geometry* at the per-Gaussian ray integral and frustum-association levels (Sec. A) while retaining *high efficiency comparable to splatting* for generic cameras (both perspective and a variety of fisheye projection models). The complete renderer still follows the standard depth ordering, independent Gaussian contributions, and alpha-compositing assumptions used by Gaussian radiance-field methods; within that pipeline, 3DGEER removes the local 2D projection approximation that dominates under wide FoV. The method contributes a unified rendering and association framework. (i) We revisit the heuristic maximum response function (Keselman & Hebert, 2022) from first principles, identify its closed-form ray-integral interpretation, and derive a numerically stable differentiable rendering formulation. (ii) We replace approximate image-space association with an exact frustum-level problem: Camera Sub-frustums (CSFs) are associated with Particle Bounding Frustums (PBFs), in direct analogy to the tile-AABB mapping that makes 3DGS efficient. For each 3D Gaussian, we derive a closed-form PBF that tightly bounds the particle and enables efficient ray-particle association without sacrificing projective correctness. (iii) We introduce Bipolar Equiangular Projection (BEAP), a supervision space that samples rays in angular coordinates, aligns image-space tiling with the underlying CSF parameterization, and empirically improves reconstruction quality over conventional pinhole or equidistant projections.

Extensive experiments on fisheye and pinhole datasets—including ScanNet++ (Yeshwanth et al., 2023), Zip-NeRF (Barron et al., 2023), Aria (Lv et al., 2025), and MipNeRF (Barron et al., 2021)—show that 3DGEER consistently outperforms prior and concurrent methods in the reported metrics. The gains are largest in the regimes for which projective correctness is most consequential: 3DGEER improves PSNR by 0.9–4.8 under wide-FoV conditions, while still obtaining 0.3–2.0 gains on pinhole views. The method also surpasses splat-based baselines on wide-FoV cameras when trained only with narrow-FoV data, demonstrating that the projective-exact formulation supports strong generalization to extremely wide-FoV rendering within the evaluated settings.

Table 1: **Summary of Particle Rendering Methods w.r.t. Projective Exactness.** The top block groups splatting-based renderers, whereas the bottom block compares ray-based methods. Several ray-based systems still use projectively approximated association, such as EWA or the Unscented Transform; among the listed methods, 3DGEER is the only one that combines ray rendering, frustum-level association, large-FoV support, high FPS, and projective-exact treatment of 3D Gaussians.

Method	Render	Particle Asso. Method	Scope	Particle Type	Large FoV	High FPS	Exact Meth.
3DGS [Kerbl et al. 2023]	Splat.	EWA	Img.	3D Gauss.	✗	✓	✗
FisheyeGS [Liao et al. 2024]	Splat.	EWA	Img.	3D Gauss.	✓	✓	✗
GS++ [Huang et al. 2024b]	Splat.	EWA	Img.	3D Gauss.	✗	✓	✗
2DGS [Huang et al. 2024a]	Ray	Splat Bound.	Img.	2D Gauss.	✗	✓	✓
GOF [Yu et al. 2024b]	Ray	EWA	Img.	3D Gauss.	✗	✗	✗
EVER [Mai et al. 2025]	Ray	BVH	Scene	3D Ellip.	✓	✗	✓
3DGRT [Moënné-Loccoz et al. 2024]	Ray	BVH	Scene	3D Gauss.	✓	✗	✓
3DGUT [Wu et al. 2025]	Ray	UT	Img.	3D Gauss.	✓	✓	✗
HTGS [Hahlbohm et al. 2025]	Ray	Splat Bound.	Img.	3D Gauss.	✗	✓	✓
3DGEER (Ours)	Ray	PBF	Frust.	3D Gauss.	✓	✓	✓

1 RELATED WORKS

1.1 SPLATTING-BASED GAUSSIAN RENDERING

Most 3DGS variants [Yu et al. 2024a, Zhang et al. 2024, Rota Bulò et al. 2024, Gao et al. 2025, Zhao et al. 2025, Kerbl et al. 2024, Hou et al. 2025, Liu et al. 2025] inherit the EWA splatting strategy [Zwicker et al. 2001, Ren et al. 2002]: each 3D Gaussian is projected to the image plane and approximated by a 2D conic. To extend this mechanism to distorted large-FoV cameras, FisheyeGS [Liao et al. 2024] and GS++ [Huang et al. 2024b] estimate Jacobians for equidistant or spherical projections and use them to update the projected 2D mean and covariance. These methods broaden splatting beyond perspective cameras, but the rendering model remains a first-order Taylor approximation to a highly nonlinear projection. Under wide-FoV distortion, the neglected higher-order terms become visually and quantitatively consequential, producing approximation error that directly degrades reconstruction quality.

The underlying limitation is geometric rather than implementation-specific: the projection of a 3D Gaussian through a nonlinear camera model, including ideal pinhole and distorted projections, is not generally a symmetric 2D Gaussian. Consequently, a renderer that linearizes the projective transformation inherits approximation error that cannot be removed by local covariance adjustment alone. 3DGEER instead evaluates Gaussian density along rays in a volumetric formulation, thereby eliminating the 2D projection approximation from per-Gaussian transmittance computation.

1.2 RAY-BASED GAUSSIAN RENDERING

Keselman & Hebert [2022] introduced the maximum-intensity response assumption for ray-Gaussian interaction. 3DGRT [Moënné-Loccoz et al. 2024] builds a 3D Gaussian ray-tracing framework on this heuristic, enabling sampling directly in ray space and reducing wide-FoV projection artifacts without explicitly splatting Gaussians to the image plane. Its main limitation is efficiency: ray-particle association depends on scene-level BVH spatial partitioning, whose traversal cost leads to low frame rates.

EVER [Mai et al. 2025] obtains closed-form transmittance-weighted ray integration by replacing Gaussians with constant-density ellipsoids. This substitution simplifies integration but reduces representational expressiveness relative to full 3D Gaussians, and the method remains runtime-limited by BVH-based association. 2DGS [Huang et al. 2024a] replaces 3D Gaussians with 2D surfels, which permits exact ray-surface intersection. HTGS [Hahlbohm et al. 2025] restores full Gaussian expressions while retaining the 2DGS ray-splat association mechanism; because this association is still screen-space constrained, it does not guarantee projective correctness for wider FoVs. 3DGEER differs by deriving closed-form bounds and a complete rendering pipeline in a camera-agnostic angular domain (Sec. D); Sec. D.3 further analyzes its computational relation to the parameterizations and rendering procedures of 2DGS and HTGS.

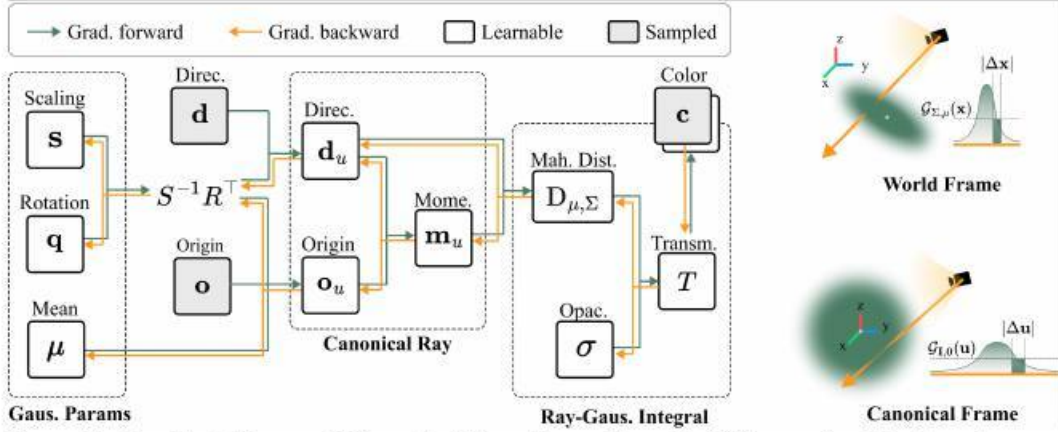


Figure 2: **Gradient Flow and Canonical Transformation.** (Left) Forward and backward propagation from the Gaussian scale s , rotation q , mean μ , and opacity coefficient σ to transmittance. (Right) Ray integral of 3D Gaussian density, with the green box indicating the density–segment-length product used in Eq. B.5

3DGUT (Wu et al. 2025), a follow-up to 3DGRT, improves runtime efficiency by adopting splatting-based particle association, achieving frame rates comparable to 3DGS. However, its acceleration relies on the Unscented Transform (UT), a sampling-based approximation used to estimate 2D conics in projective image space. A similar limitation exists in GOF (Yu et al. 2024b), which directly applies the EWA trick to fit 2D bounding conics and projective centers for association. Notably, under extreme-FoV conditions, imprecise AABB estimation and the resulting inaccurate tile-to-AABB associations can lead to visible grid-line artifacts (see Fig. 1 Left).

In contrast to these methods, our approach solves ray–particle association directly as a frustum-level 3D problem (Fig. 1 Right), preserving the projective correctness of ray-based rendering across arbitrary camera models while maintaining rendering speeds comparable to 3DGS. Tab. 1 summarizes representative methods that address projective approximation errors and highlights the specific combination of rendering, association, FoV support, and efficiency targeted by 3DGEER.

2 METHOD

3DGEER is a Gaussian rendering framework designed to preserve projective geometry within the standard alpha-compositing formulation while retaining real-time throughput for arbitrary-FoV cameras. It addresses the two bottlenecks that separate existing renderers—*projection approximation* in splatting-based pipelines and *BVH overhead* in ray-based pipelines—through three coupled components:

- Ray-based rendering formulation (Sec. 2.1): a closed-form differentiable integral of Gaussian density along 3D rays that removes the local projection linearization used by splatting.
- Particle Bounding Frustum (PBF) (Sec. 2.2): a ray–particle association scheme that replaces BVH traversal with closed-form frustum–particle bounds, coupling geometric accuracy with efficient parallel execution.
- Bipolar Equiangular Projection (BEAP) (Sec. 2.3): an angular image representation for arbitrary-FoV camera models that aligns supervision and tiling with PBF association.

The Appendix supplies the supporting derivations: (i) preliminaries and notation (Sec. B.1), (ii) the proof behind Sec. 2.1 connecting maximum response to the per-Gaussian ray integral (Sec. B), (iii) backward-gradient derivations and numerical-stability analysis (Sec. C), and (iv) the derivation of geometrically exact association (Sec. D).

2.1 RENDERING WITH EXACT PROJECTIVE GEOMETRY

A projectively exact per-Gaussian renderer requires a closed-form expression for the density accumulated along each camera ray, independent of any local image-plane approximation. We obtain

this expression by mapping each anisotropic world-space Gaussian to a canonical coordinate system in which the Gaussian becomes isotropic. Under this transformation, the ray integral reduces to a measure-preserving change of variables and admits a compact closed-form solution.

Canonical transformation. For each Gaussian with mean μ , rotation R , and scaling S , we define the mapping

$$\begin{bmatrix} \mathbf{x} \\ 1 \end{bmatrix} = \begin{bmatrix} RS & \mu \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{u} \\ 1 \end{bmatrix}, \quad (1)$$

which maps the canonical coordinate $\mathbf{u}$ to the world coordinate $\mathbf{x}$. The relation $\Delta \mathbf{x} = RS \Delta \mathbf{u}$ incorporates the Jacobian determinant into the density measure (Fig. 2 Right), so every Gaussian can be expressed in the shared isotropic form

$$\mathcal{G}_{\mathbf{I},\mathbf{o}}(\mathbf{u}) = \frac{1}{\rho} \exp\left(-\frac{1}{2}\|\mathbf{u}\|^2\right). \quad (2)$$

Closed-form transmittance. For a ray $\mathbf{r}(t) = \mathbf{o} + t\mathbf{d}$, its canonical form is $\mathbf{r}_u(t) = \mathbf{o}_u + t\mathbf{d}_u$ with

$$\mathbf{o}_u = S^{-1}R^\top(\mathbf{o} - \mu), \quad \mathbf{d}_u = S^{-1}R^\top\mathbf{d}. \quad (3)$$

The corresponding per-Gaussian accumulated transmittance is

$$T(\mathbf{o}, \mathbf{d}) = \sigma \int_{t \in \mathbb{R}} \mathcal{G}_{\mathbf{I},\mathbf{o}}(\mathbf{o}_u + t\mathbf{d}_u) dt = \sigma \exp\left(-\frac{1}{2}D_{\mu,\Sigma}^2(\mathbf{o}, \mathbf{d})\right), \quad (4)$$

where

$$D_{\mu,\Sigma}^2(\mathbf{o}, \mathbf{d}) = \frac{\|\mathbf{o}_u \times \mathbf{d}_u\|^2}{\|\mathbf{d}_u\|^2} \quad (5)$$

is the squared perpendicular Mahalanobis distance between the ray and the Gaussian center.

Discussion. This expression shows that the per-Gaussian transmittance is determined solely by the ray–Gaussian distance in canonical space, giving an efficient closed form that remains exact with respect to the ray’s projective geometry. It also clarifies the “maximum response” heuristic used in prior work (Keselman & Hebert, 2022; Moënné-Loccoz et al., 2024): the heuristic coincides with the derived integral, and Sec. B makes its projective-exact interpretation explicit. Because the same derivation yields the backward gradient chain (Fig. 2 Left; Sec. C), 3DGEER avoids the numerical instabilities reported in prior ray-based work (Yu et al., 2024b) and removes redundant post-processing such as degenerate-Gaussian filtering.

2.2 RAY-PARTICLE ASSOCIATION WITH EXACT PROJECTIVE GEOMETRY

In ray-based Gaussian rendering, both reconstruction accuracy—especially for large-FoV cameras—and runtime throughput are governed by ray–particle association. Instead of testing individual rays against scene-level acceleration structures, 3DGEER formulates association as a frustum-intersection problem between Camera Sub-Frustums (CSFs) and Particle Bounding Frustums (PBFs). A PBF tightly encloses the contribution region of a Gaussian at a prescribed distance from its center (Fig. 3); a Gaussian is assigned to a CSF when the PBF intersects that CSF. The associated Gaussians are then evaluated for every ray inside the CSF, preserving the closed-form per-ray integral while enabling tile-like parallel scheduling.

To compute a PBF efficiently, we parameterize camera-space ray directions by two angular coordinates (Fig. E.1):

$$\theta = \arctan\left(\frac{\mathbf{d}_{c,x}}{\mathbf{d}_{c,z}}\right), \quad \phi = \arctan\left(\frac{\mathbf{d}_{c,y}}{\mathbf{d}_{c,z}}\right), \quad (6)$$

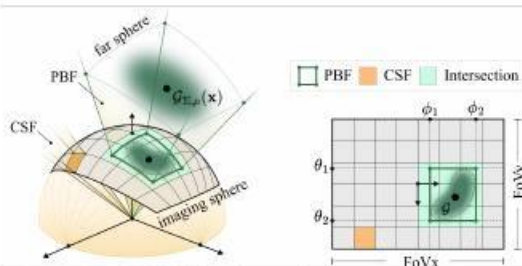


Figure 3: **PBF-CSF Association.** (Left) A PBF is defined by four angular planes tangent to the 3D Gaussian ellipsoid. (Right) The PBF–CSF intersection is unfolded onto the BEAP imaging plane, enabling tile-level association without image-plane conic approximation.

where $(\mathbf{d}_{c,x}, \mathbf{d}_{c,y}, \mathbf{d}_{c,z})$ is the normalized direction vector from the optical center to the camera-space point $\mathbf{d}_c$. The angles θ and ϕ are the horizontal (xz) and vertical (yz) incidence angles of the ray, each lying in $-\frac{\pi}{2}$ to $\frac{\pi}{2}$. A PBF is therefore specified by four angular planes, two bounding θ and two bounding ϕ (Fig. 3):

$$\mathbf{g}_{\theta_{1,2}} = (-1, 0, \tan \theta_{1,2}, 0)^\top, \quad \mathbf{g}_{\phi_{1,2}} = (0, -1, \tan \phi_{1,2}, 0)^\top, \quad (7)$$

where $\theta_1 < \theta < \theta_2$ and $\phi_1 < \phi < \phi_2$ define the angular support of the frustum.

For a camera extrinsic matrix $V = [R_c | \mathbf{t}_c] \in \text{SE}(3)$ and a canonical-to-world transformation H (Eq. B.4), the angular planes defined in view space are transformed into canonical space by the inverse-transpose action of the camera-to-canonical transform $(VH)^{-1}$. This transformation yields canonical plane constraints from which the PBF bounds are solved. For the θ bounds, the transformed plane is

$$\mathbf{g}_{u_{1,2}} = (VH)^\top \mathbf{g}_{\theta_{1,2}} = -(VH)_0 + \tan \theta_{1,2} (VH)_2, \quad (8)$$

where $(VH)_i$ denotes the i -th row of the matrix, written as a column vector.

For a Gaussian ellipsoid defined by a λ -standard-deviation contour, the local angular bounds are obtained by enforcing tangency between the transformed planes and the ellipsoidal contour. This tangency condition gives the quadratic constraint

$$((\lambda^2, \lambda^2, \lambda^2, -1) \circ \mathbf{g}_u^\top) \cdot \mathbf{g}_u = 0, \quad (9)$$

where $\circ$ denotes the Hadamard (element-wise) product, and λ controls the contour spread and is typically set to 3. The same constraint is applied to $\mathbf{g}_v^\top$ to recover $\phi_{1,2}$. The roots of these quadratics give the tangent values of $[\theta_1, \theta_2]$ and $[\phi_1, \phi_2]$, which define the angular limits of the PBF. Because the quadratic is solved in canonical space, the resulting bounds depend on the true camera-space 3D covariance Σ_c and mean μ_c rather than on an intermediate 2D conic. For the θ bounds, $c = \tan \theta_{1,2}$ satisfies

$$\mathcal{T}_{22}c^2 - 2\mathcal{T}_{02}c + \mathcal{T}_{00} = 0, \quad \text{where} \quad \mathcal{T}_{ij} = \lambda^2 \Sigma_c^{i,j} - (\mu_c \mu_c^\top)^{i,j}. \quad (10)$$

The resulting solver is closed form and efficient (Sec. D.1). It removes the intermediate conic approximations used by EWA and UT and avoids the screen-space constraints in HTGS and 2DGS that restrict their FoV applicability.

2.3 BIPOLAR EQUIANGULAR PROJECTION (BEAP)

BEAP provides the angular image representation used by the proposed renderer. It represents large-FoV images without FoV loss, assigns image tiles and their corresponding CSFs the same two- θ /two- ϕ parameterization, and thereby supports efficient PBF association with favorable GPU parallelism. Its ray samples are also more uniformly distributed in 3D space than those produced by conventional projective rendering, which empirically improves training efficiency and reconstruction quality in our ablations.

Concretely, rays are sampled uniformly in spherical angular coordinates (θ, ϕ) (Eq. 6) and then mapped to the discrete $w \times h$ image lattice by a linear scaling and center shift:

$$(x, y) = \left\lfloor \frac{w\theta}{\text{FoV}_x} + \frac{w+1}{2}, \frac{h\phi}{\text{FoV}_y} + \frac{h+1}{2} \right\rfloor, \quad (11)$$

where $\text{FoV}_x, \text{FoV}_y$ denote the horizontal and vertical fields of view. As Fig. 4 shows, uniform sampling in (θ, ϕ) yields more balanced coverage of the view frustum. This differs from projective sampling, which over-allocates samples to peripheral regions, and from equidistant projection, which oversamples the image center; BEAP is therefore positioned not as a replacement for camera calibration, but as a supervision and tiling space matched to the renderer's angular association geometry.

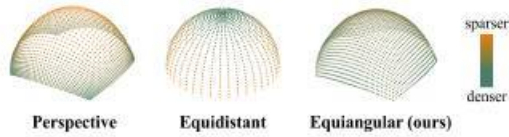


Figure 4: **Comparison of Ray Distributions under Varying Projections.** Pixels from three imaging spaces are projected onto the unit sphere. BEAP provides the most uniform angular coverage among the compared projections, aligning supervision with the frustum parameterization used by PBF association.

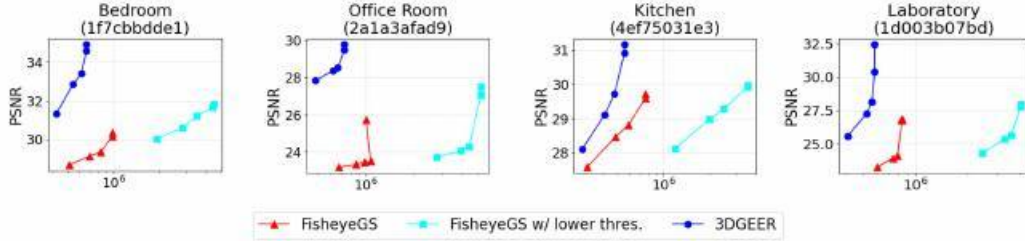


Figure 5: **PSNR $\uparrow$ Trends When Scaling-Up Gaussian Counts.** Increasing the Gaussian count in a splatting-based fisheye renderer does not close the projective-exactness gap; full metrics appear in Fig. K.3 and Tab. K.3



Figure 6: **Extreme-FoV Generalization.** Under wider-FoV evaluation, 3DGEER reconstructs complete scenes with fewer Gaussians, whereas FisheyeGS exhibits severe peripheral quality degradation.

3 EXPERIMENTS

We evaluate 3DGEER across complementary pinhole and fisheye benchmarks to test whether projective exactness improves both wide-FoV reconstruction and standard novel-view synthesis while preserving real-time efficiency. Unless otherwise stated, experiments use an RTX 4090 GPU with 64GB of RAM; implementation details are given in Sec. G

Datasets. For generic distorted cameras, where projective approximation is most consequential, we use all scenes from Fisheye ZipNeRF (Barron et al., 2023), five ScanNet++ scenes (Yeshwanth et al., 2023), and two Aria scenes (Lv et al., 2025). ZipNeRF and ScanNet++ provide 180° diagonal FoV views, whereas Aria provides a 110° circular FoV. For conventional pinhole evaluation, we use Pinhole ZipNeRF and MipNeRF360 (Barron et al., 2021).

Evaluation Metrics and Protocol. We report PSNR $\uparrow$, SSIM $\uparrow$, LPIPS $\downarrow$, runtime performance (FPS $\uparrow$), and Gaussian counts $\downarrow$. To avoid bias from different internal image representations, all outputs are projected back to the image space and evaluated over the full FoV after 30k training iterations.

3.1 SCALING GAUSSIANS VS. PROJECTIVE EXACTNESS IN WIDE-FOV CAMERAS

(Celarek et al., 2025) argues that increasing the number of Gaussians, up to 1M, can mitigate the linear-approximation gap between splat-based and ray-based methods. That evidence is limited to small object-centric scenes and does not test the stronger distortions introduced by wide-FoV cameras. We therefore stress-test the scaling hypothesis on ScanNet++ by lowering the densification threshold of the splat-based fisheye method FisheyeGS, producing 3.6–5M Gaussians on an H200.

Fig. 5 shows a persistent PSNR separation between FisheyeGS (red) and 3DGEER (blue) at matched Gaussian counts from 4k to 30k iterations, after which densification typically saturates. The scaled FisheyeGS variant (light blue) substantially increases memory and computation, yet its performance

Table 2: **Comparison on the ScanNet++ Dataset.** Methods are trained on either the full FoV or the central region and evaluated on full, central, and peripheral regions to isolate performance under different distortion levels. Region-wise metrics are weighted by pixel masks; LPIPS is omitted from regional splits because it is not directly separable by region. Full statistics are reported in Tab. K.2

Method	Training FoV	Training Space	Full FoV			Central		Peripheral	
			PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$
3DGS	Central	Persp.	-	-	-	31.26	0.948	-	-
FisheyeGS	Full	Eq.	27.81	0.946	0.139	32.44	0.956	23.28	0.914
EVER	Full	KB	29.47	0.924	0.167	29.93	0.924	28.72	0.925
3DGUT	Full	Eq.	30.64	0.944	0.150	31.87	0.945	28.84	0.937
3DGEER (Ours)	Central	BEAP	29.93	0.949	0.130	32.86	0.954	26.64	0.930
3DGEER (Ours)	Full	BEAP	31.50	0.953	0.126	32.64	0.955	28.94	0.945

Table 3: **Comparison on the ZipNeRF Dataset.** The cross-camera protocol trains each method on either Fisheye (FE, 1/8 resolution) or Pinhole (PH, 1/4 resolution) data and evaluates it on both Fisheye and Pinhole test sets; full per-scene statistics are provided in Tab. K.9

Train Data Test Data Method	1/8 FE		1/8 FE		1/8 FE		1/4 PH		1/4 PH		1/4 PH	
	FE	PH	FE	PH	FE	PH	FE	PH	FE	PH	FE	PH
	PSNR $\uparrow$		SSIM $\uparrow$		LPIPS $\downarrow$		PSNR $\uparrow$		SSIM $\uparrow$		LPIPS $\downarrow$	
FisheyeGS	23.18	26.44	0.858	0.868	0.211	0.239	19.43	26.61	0.791	0.889	0.247	0.212
EVER	25.17	26.14	0.880	0.851	0.153	0.237	22.90	26.45	0.835	0.862	0.207	0.222
3DGUT	24.77	25.59	0.879	0.804	0.183	0.324	18.61	25.96	0.662	0.841	0.300	0.270
3DGEER (Ours)	26.24	27.62	0.897	0.888	0.140	0.214	23.39	27.61	0.846	0.863	0.209	0.254

saturates at 29.3 PSNR; 3DGEER reaches 32.1 PSNR with only 550–700K Gaussians. These results indicate that, for the evaluated wide-FoV setting, particle scaling alone does not remove the projective-exactness gap. Full statistics are reported in Tab. K.3

3.2 COMPREHENSIVE EVALUATION ON ALL-FOV NOVEL VIEW SYNTHESIS

We next compare 3DGEER with representative baselines on fisheye and pinhole novel-view-synthesis benchmarks. The evaluation couples aggregate metrics with dataset-specific analyses so that the effect of projective exactness is visible in both wide-FoV and standard settings.

ScanNet++. Tab. 2 shows that 3DGEER achieves the strongest full-FoV performance among the evaluated methods, with 31.50 PSNR, 0.953 SSIM, and 0.126 LPIPS. To expose the effect of distortion, we additionally evaluate central and peripheral regions (Sec. G). Baselines remain competitive near the image center but lose fidelity in the periphery, whereas 3DGEER maintains high quality across both regions and produces sharper boundaries with fewer artifacts (Fig. K.4). Even when trained only on central crops, 3DGEER still exceeds FisheyeGS across all reported regions, supporting the role of projective-exact rendering beyond direct full-FoV supervision.

Fig. 6 further evaluates extreme-FoV generalization by training on the original dataset FoV (180° diagonal FoV) and testing under a wider circular 180° FoV. 3DGEER reconstructs complete scenes with fewer Gaussians, while splatting-based methods such as FisheyeGS degrade markedly in peripheral regions; additional extreme-FoV visualizations are provided in Fig. K.1–K.2

ZipNeRF. ZipNeRF provides frame-wise aligned pinhole and fisheye views, enabling a direct *cross-camera generalization* protocol: methods are trained on fisheye and tested on pinhole, or trained on pinhole and tested on fisheye. Tab. 3 shows that 3DGEER attains the best PSNR and SSIM in every train–test pairing and the best or near-best LPIPS values. When trained on fisheye views, it benefits from large-FoV supervision through geometrically exact ren-

Table 4: **Quantitative Results on the MipNeRF360 Dataset.** In addition to quality metrics, FPS is reported using an RTX 4090 as the default benchmark. Numbers sourced from original works are marked with $\dagger$ (measured on RTX 6000 Ada) and $\ddagger$ (measured on RTX 5090).

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FPS $\uparrow$
3DGS	27.21	0.815	0.214	343
EVER	27.51	0.825	0.233	36
3DGRT	27.20	0.818	0.248	52 $\dagger$ / 68 $\ddagger$
3DGUT	27.26	0.810	0.218	265 $\dagger$ / 317 $\ddagger$
3DGEER (Ours)	27.76	0.821	0.210	327

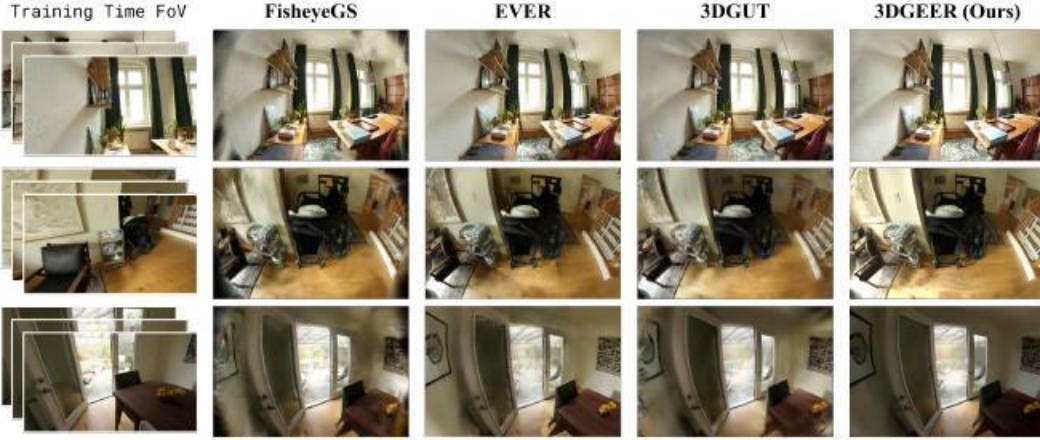


Figure 7: **Qualitative Results on the ZipNeRF Dataset with Cross-Camera Generalization.** Differences are most visible in the challenging Pinhole (PH) train–Fisheye (FE) test setting, where 3DGEER preserves detail and suppresses peripheral artifacts more effectively than the baselines.

Table 5: **Training-Time Transmittance Replacement on ScanNet++.** Using splatting-based transmittance increases the Gaussian count due to larger approximation error, yet the perceptual gap remains compared with our projective-exact formulation (Tab. K.7 shows the full stats).

Transmittance Method	Full FoV			Central		Peripheral		Avg. Gaussian Number (k) ↓
	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	PSNR ↑	SSIM ↑	
3DGS Splats	22.86	0.878	0.177	27.16	0.884	18.56	0.855	1396.4
FisheyeGS Splats	27.90	0.948	0.141	32.49	0.957	23.36	0.917	920.6
Projective-Exact (Ours)	31.50	0.953	0.126	32.64	0.955	28.94	0.945	591.8

dering; when trained on pinhole views, it still improves PSNR on both fisheye and pinhole tests. FisheyeGS remains strongest only in the pinhole-to-pinhole LPIPS/SSIM entries, whereas 3DGUT drops sharply under pinhole-to-fisheye transfer. EVER and 3DGEER generalize better because they preserve geometric exactness, and 3DGEER is consistently superior in this comparison. Qualitatively, Fig. 7 shows finer detail recovery and fewer peripheral artifacts in the most challenging pinhole-to-fisheye setting.

On **MipNeRF360**, 3DGEER also performs strongly under standard narrow-FoV conditions. Tab. 4 reports the best PSNR and LPIPS among the listed methods and competitive SSIM, while maintaining 327 FPS on the RTX 4090. This makes 3DGEER more than $5\times$ faster than exact ray-marching/tracing methods such as EVER and 3DGRT and the only projective-exact method with runtime comparable to 3DGS. It also exceeds the projective-approximation baselines 3DGS and 3DGUT in PSNR, SSIM, and LPIPS. Additional quantitative results on **Aria**, an egocentric fisheye dataset with moderate 110° circular FoV and low-light imaging, together with further visualizations, are provided in Sec. J.

3.3 COMPREHENSIVE RUNTIME ANALYSIS

We analyze runtime on RTX 4090 against optimized splatting-based pipelines: 3DGS on **MipNeRF360** and FisheyeGS on **ScanNet++**. Although ray-based evaluation makes the rendering stage slightly more expensive, PBF association substantially reduces Gaussian duplication and sorting overhead (Tab. K.4), yielding end-to-end efficiency comparable to splatting while preserving projective exactness.

PBF produces tight bounds: each tile is associated with only ~ 475 Gaussians on average, which is $3\text{--}5\times$ fewer than EWA or UT (Tab. K.6). This reduction accounts for the $2.5\text{--}5\times$ speedup in associa-

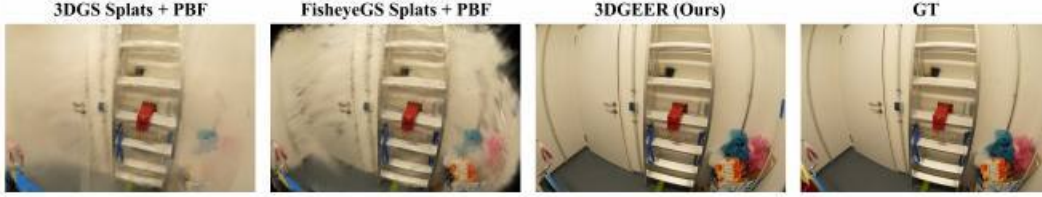


Figure 8: **Rendering-Time Transmittance Func. Replacement.** Replacing projective-exact transmittance with the splatting approximation only at test time leads to mismatched culling.

Table 6: **BEAP vs. Alternative Supervision Schemes.** BEAP consistently outperforms other projections across all metrics on ScanNet++. (Tab. K.8 shows full stats).

Method	Training FoV	Training Space	Full FoV			Central		Peripheral	
			PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$
3DGEER w/o BEAP	Central	Persp.	29.84	0.943	0.131	32.21	0.951	26.23	0.915
3DGEER w/o BEAP	Full	Persp.	21.11	0.853	0.300	21.32	0.855	20.46	0.846
3DGEER w/o BEAP	Full	Eq.	31.05	0.948	0.135	32.21	0.952	28.56	0.935
3DGEER (Ours)	Full	BEAP	31.50	0.953	0.126	32.64	0.955	28.94	0.945

tion (Tab. K.5). Projection-based EWA and UT association conservatively overestimate screen-space extent, causing overdraw and inefficient memory access; even with SnugBox acceleration (Hanson et al. 2025), they remain affected by miscentered density approximations and can intensify artifacts (Fig. 1). PBF therefore contributes both efficiency and geometric reliability: Tab. K.6 shows that, even when ray-based rendering Gaussian fields trained with other association schemes, PBF remains artifact-free.

3.4 LOCAL AFFINE APPROXIMATION VS. PROJECTIVE-EXACT INTEGRAL

To isolate the effect of our projective-exact transmittance formulation (Eq. 4), we replace it with the commonly used local affine splats (i.e., 2D conic as approximated in 3DGS and FisheyeGS).

Rendering-Time Approx. To better highlight the difference between the two rendering formulations, we replace the transmittance with the splatting-based approximation only at test time, while keeping all projective-exact components during training. As shown in Fig. 8 when the transmittance relies on the splat approximation but the associations are computed using our exact projective geometry, their inconsistency leads to mismatched culling. This experiment magnifies and explains the gridline artifact reported in 3DGUT (Fig. 1), where the mismatch arises in the opposite direction—ray-based transmittance combined with conic-approximate association.

Training-Time Approx. As shown in Tab. 5 experimented on ScanNet++, when splatting-based transmittance is also used during training while association remains projective-exact, the optimizer and densification partially compensate for the approximation error and the cross-view inconsistency introduced by splatting. The larger loss induces stronger gradients throughout training, ultimately leading to a larger ($1.6\text{--}2.4\times$) number of Gaussians being added to fit the scene. However, despite this increased Gaussian count, the perceptual gap introduced by the approximation remains, indicating that the error cannot be fully corrected through densification alone.

3.5 BEAP ON/OFF STUDY

We further compare BEAP supervision against other widely used projection models on ScanNet++. In this experiment, we keep the projective-exact integral and PBF for Gaussian transmittance and association fixed, and vary only the ray-sampling distribution used for supervision (pinhole, equidistant, and BEAP). As shown in Tab. 6 and Fig. K.5, our BEAP-based supervision outperforms all alternatives across every metric. Its more uniform sampling provides smoother and more consistent gradients, enabling better convergence on fine details.

The discrepancy becomes more pronounced under pinhole cameras. Under a fixed resolution, a well-balanced information distribution can matter more than simply using a larger FoV. Notably,

when training in the perspective domain, the full FoV can even underperform the central region (see Tab. 6 rows 1–2) because large-FoV images (Fig. K.2 Left) have highly non-uniform angular sampling: informative content is concentrated at the center, while the less informative periphery occupies most of the image area and becomes disproportionately oversampled.

4 CONCLUSIONS

We presented 3DGEER, a complete, closed-form solution for projective-exact volumetric Gaussian rendering derived from first principles. To complement this formulation, we introduced an efficient PBF computation method for ray-particle association, enabling high-speed rendering while maintaining exactness in projective geometry. Additionally, we proposed the BEAP image representation to facilitate effective color supervision under wide-FoV camera models. Extensive experiments demonstrate that 3DGEER consistently outperforms state-of-the-art methods across multiple datasets, with particularly strong results in challenging wide-FoV scenarios.

ETHICS STATEMENT

This work builds on publicly available datasets (ScanNet++, ZipNeRF, Aria, and MipNeRF) and does not involve any human subjects, private data, or personally identifiable information. The methods developed are intended for advancing 3D scene reconstruction and rendering research, with potential applications in XR, robotics and autonomous driving. We do not foresee direct harmful consequences, but we note that any misuse of high-fidelity 3D reconstruction techniques could raise privacy concerns. To mitigate this, we restrict our experiments to research datasets that are widely used in the vision community.

REPRODUCIBILITY STATEMENT

We make significant efforts to ensure reproducibility. All mathematical derivations and proofs are provided in the Appendix (Secs. B, C and D). The datasets used in our experiments are publicly available (ScanNet++, ZipNeRF, Aria, and MipNeRF). Details of preprocessing, training protocols, and evaluation metrics are described in Sec. G. Our implementation will be released with training and evaluation scripts upon publication to further facilitate reproducibility.

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810	CONTENTS	
811		
812	1 Related Works	3
813		
814	1.1 Splatting-Based Gaussian Rendering	3
815	1.2 Ray-Based Gaussian Rendering	3
816		
817	2 Method	4
818		
819	2.1 Rendering with Exact Projective Geometry	4
820	2.2 Ray-Particle Association with Exact Projective Geometry	5
821	2.3 Bipolar Equiangular Projection (BEAP)	6
822		
823		
824	3 Experiments	7
825		
826	3.1 Scaling Gaussians vs. Projective Exactness in Wide-FoV Cameras	7
827	3.2 Comprehensive Evaluation on All-FoV Novel View Synthesis	8
828	3.3 Comprehensive Runtime Analysis	9
829	3.4 Local Affine Approximation vs. Projective-Exact Integral	10
830	3.5 BEAP On/Off Study	10
831		
832		
833	4 Conclusions	11
834		
835		
836	a Ethics Statement	12
837		
838	b Reproducibility Statement	12
839		
840	c LLM Usage Statement	18
841		
842		
843	A Appendix: Exactness and Limitation	18
844		
845	B Appendix: Ray-Gaussian Transmittance with Projective Exactness	18
846		
847	B.1 Preliminaries And Notations	18
848	B.2 Differentiable Rendering with Projective Exactness	19
849	B.3 Ray Integral of Isotropic 3D Gaussian	20
850		
851		
852	C Appendix: Cookbook of Gradient Computation	21
853		
854	C.1 Chain Rule	21
855	C.2 Two-Stage Propagation	21
856	C.3 Gradients for Canonical Ray	21
857	C.4 Gradients for Whitening Matrix	22
858	C.5 Gradients for Scaling and Rotation	22
859	C.6 Gradients for 3D Mean	22
860	C.7 Numerical Stability of Our Differentiable Framework	22
861		
862		
863	D Appendix: Additional Details in Association	23

864	D.1 Particle Bounding Frustum (PBF) Solver	23
865	D.2 Details in Computing CSF-PBF Intersection	24
866	D.3 Relation to Prior Works	25
867		
868		
869	E Appendix: BEAP Image Inverse Interpolation	26
870		
871	Pinhole Camera Model	26
872	KB Fisheye Model	26
873		
874	F Appendix: High-level Algorithmic Summary	27
875		
876		
877	G Appendix: Additional Implementation Details	27
878	Baselines	27
879	Implementation Details	27
880	Central and peripheral regions in ScanNet++ evaluation	28
881	Cross-camera generalization evaluation on ZipNeRF	28
882		
883		
884	H Appendix: EWA Splatting Preliminaries	28
885		
886	From camera coordinates to ray coordinates	28
887	Local affine approximation	28
888	Splatting approximation error	29
889		
890		
891	I Appendix: Advantage over Prior Ray-Tracing Methods	29
892		
893	J Appendix: Results on Egocentric Dataset	30
894		
895	K Appendix: Additional Material and Results	31
896		
897	K.1 Additional Material.	31
898	K.2 Additional Results.	31
899		
900		
901		
902		
903		
904		
905		
906		
907		
908		
909		
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911		
912		
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LLM USAGE STATEMENT

We used a large language model (ChatGPT, OpenAI) as a writing assistant. Specifically, the LLM was employed to (i) improve the clarity and readability of the manuscript, (ii) suggest stylistic edits to figure captions and section transitions, and (iii) provide alternative phrasings for technical descriptions without altering the scientific content.

The LLM was not involved in research ideation, algorithm design, data analysis, or experimental validation. All technical contributions, mathematical derivations, and results are the work of the authors. The authors take full responsibility for the content of this paper.

A APPENDIX: EXACTNESS AND LIMITATION

We clarify that our exactness refers to *eliminating linear approximation in projective geometry* during both the “rendering” and “ray-particle association” procedure, where errors will not decrease as the number of Gaussians increases under our settings (see Sec. 3.1). We do not claim exactness in the full physical volumetric sense. Our framework still involves approximations like ordering, self-attenuation, and overlapping which are our limitations.

Limitations. Our formulation integrates each Gaussian independently; hence it does not explicitly resolve self-occlusion, and ordering is handled through depth-based sorting. As also shown in the supplementary videos, this may lead to occasional popping artifacts. To our knowledge, EVER (Mai et al. 2025) is the only method that analytically resolves ordering and self-occlusion by approximating each Gaussian as a constant-density ellipsoid, though this comes with a substantial reduction in rendering speed.

Importantly, prior ray-tracing-based Gaussian renderers such as Fuzzy Metaballs (Keselman & Hebert 2022) and subsequent methods including 3DGR/UT (Moënné-Loccoz et al. 2024; Wu et al. 2025) rely on the “maximum-response” assumption for ray-Gaussian interaction. This assumption inherently ignores ordering and self-attenuation, yet has been shown to introduce negligible error in practice (see Celarek et al. (2025) Fig. 9-11 / Sec. 7.1) regardless of the number of Gaussians. This explains that our depth-based ordering still can achieve the same practical accuracy as existing ray-tracing-based Gaussian renderers.

Future work. While the practical impact of ordering and self-occlusion remains small in current Gaussian-based ray tracing, these factors fundamentally limit exact volumetric correctness. A promising future direction is to incorporate analytic or learnable self-attenuation models—potentially inspired by ellipsoidal integration (e.g., EVER (Mai et al. 2025)) but without incurring heavy runtime costs. Another important avenue is handling ordering in a more principled manner, which may further reduce the rare popping artifacts observed under extreme configurations. We believe these limitations are orthogonal to our contributions on projective exactness, and addressing them would make the framework even more robust for large-scale or high-dynamic scenes.

B APPENDIX: RAY-GAUSSIAN TRANSMITTANCE WITH PROJECTIVE EXACTNESS

B.1 PRELIMINARIES AND NOTATIONS

We follow the classic parameterization of 3D Gaussian primitives as introduced in prior work (Zwicker et al. 2001). Each 3D Gaussian distribution is defined by a mean position $\mu \in \mathbb{R}^3$ and a 3D covariance matrix Σ in the world coordinate system:

$$\mathcal{G}_{\Sigma, \mu}(\mathbf{x}) = \frac{1}{\rho |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (\mathbf{x} - \mu)^\top \Sigma^{-1} (\mathbf{x} - \mu) \right), \quad (\text{B.1})$$

where we adopt the normalization constant ρ as $\sqrt{2\pi}$, and $\mathbf{x} \in \mathbb{R}^3$ denotes a point in world space. This formulation differs from recent methods such as 3DGS (Kerbl et al. 2023) and 3DGUT (Wu et al. 2025), which omit the determinant term $|\Sigma|^{1/2}$ to simplify splatting or association during rendering. The form of our covariance matrix obeys: $\Sigma = R S S^\top R^\top$, given a scaling matrix

$S = \text{diag}(\mathbf{s})$ constructed from the scaling vector $\mathbf{s} \in \mathbb{R}^3$, and a rotation matrix $R \in \text{SO}(3)$ derived from a unit quaternion $\mathbf{q} \in \mathbb{R}^4$.

To evaluate the transmittance contributed by a single 3D Gaussian particle along a ray $\mathbf{r}(t) = \mathbf{o} + t\mathbf{d}$, we have:

$$T(\mathbf{o}, \mathbf{d}) = \sigma \int_{t \in \mathbb{R}} \mathcal{G}_{\Sigma, \mu}(\mathbf{o} + t\mathbf{d}) dt, \quad (\text{B.2})$$

where σ is the opacity coefficient associated with the Gaussian.

Finally, considering a set of 3D Gaussian particles sorted by depth contributing to the rendering color of the given ray, the rendered color C is computed as:

$$C(\mathbf{o}, \mathbf{d}) = \sum_i c_i(\mathbf{o}, \mu_i) T_i \prod_{j=1}^{i-1} (1 - T_j), \quad (\text{B.3})$$

where $c_i(\mathbf{o}, \mu_i)$ derives the view-dependent color of the i -th Gaussian as seen from the optical center $\mathbf{o}$, parameterized using spherical harmonics (SH) (Kerbl et al. 2023). Through minimizing the reconstruction loss between the rendered color and the observed color along each ray, the parameters $\mathbf{s}, \mathbf{q}, \mu$, the coefficient σ as well as the SH coefficients, are jointly optimized via backpropagation.

B.2 DIFFERENTIABLE RENDERING WITH PROJECTIVE EXACTNESS

The core challenge in achieving projective-exact Gaussian rendering in a ray marching formulation lies in obtaining a closed-form solution for integrating Gaussian density and color along a 3D ray. To address this, we first transform the ray—relative to each Gaussian—into a canonical coordinate system where the transmittance integral (see Eq. B.2) remains consistent and the Gaussians become isotropic. We then derive the exact ray color from first principles.

Specifically, we define a canonical coordinate system shared across all 3D Gaussian primitives. A point $\mathbf{u} \in \mathbb{R}^3$ in this canonical space is mapped to world coordinates for each Gaussian via:

$$\begin{bmatrix} \mathbf{x} \\ 1 \end{bmatrix} = H \begin{bmatrix} \mathbf{u} \\ 1 \end{bmatrix} = \begin{bmatrix} RS & \mu \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{u} \\ 1 \end{bmatrix}, \quad (\text{B.4})$$

From this transformation, an infinitesimal segment $\Delta \mathbf{u}$ along a ray in the canonical frame corresponds to a segment $\Delta \mathbf{x} = RS \Delta \mathbf{u}$ in the world frame.

To ensure that the transmittance contributed by each Gaussian along the ray is preserved across coordinate systems, we require that the elementary transmittance (i.e., the product of density and segment length) remains invariant under the transformation (see Fig. 2 Right):

$$\mathcal{G}_{\mathbf{I}, \mathbf{0}}(\mathbf{u}) |\Delta \mathbf{u}| = \mathcal{G}_{\Sigma, \mu}(\mathbf{x}) |\Delta \mathbf{x}|. \quad (\text{B.5})$$

This constraint implies that the Jacobian determinant is absorbed into the measure. Note that $|\frac{\Delta \mathbf{x}}{\Delta \mathbf{u}}| = |RS| = |\Sigma|^{1/2}$. We can thus express the corresponding canonical density as:

$$\mathcal{G}_{\mathbf{I}, \mathbf{0}}(\mathbf{u}) = \frac{1}{\rho} \exp\left(-\frac{1}{2} \mathbf{u}^\top \mathbf{u}\right), \quad (\text{B.6})$$

which is isotropic and shared across all primitives in canonical space.

This leads to an important conclusion that the exact closed-form solution for the transmittance integral of each Gaussian along the ray $\mathbf{r}(t)$ can thus be obtained by computing the accumulated density along the transformed ray $\mathbf{r}_u(t) = \mathbf{o}_u + t\mathbf{d}_u$ in the canonical isotropic Gaussian. Note that $t < 0$ is always negligible unless the optical center is within 3-sigma, then all 3D Gaussian rendering frameworks cull the Gaussian when optimization.

This admits a *closed-form* solution in terms of the perpendicular distance from the ray to the Gaussian center, as captured in Theorem 1

$$T(\mathbf{o}, \mathbf{d}) = \sigma \int_{t \in \mathbb{R}} \mathcal{G}_{\mathbf{I}, \mathbf{0}}(\mathbf{o}_u + t\mathbf{d}_u) dt = \sigma \exp\left(-\frac{1}{2} (D_{\mu, \Sigma}(\mathbf{o}, \mathbf{d}))^2\right), \quad (\text{B.7})$$

where $D_{\mu,\Sigma}(\mathbf{o}, \mathbf{d})$ denotes the perpendicular distance from the Gaussian center (i.e., the origin in the canonical space) to the transformed ray $\mathbf{r}_u(t)$. Intuitively, this quantity also represents the minimal *Mahalanobis distance* from any points on the ray to the Gaussian in world space, measuring how close the ray passes by the particle distribution.

The remaining question is just about how to compute the perpendicular distance from the canonical origin to the transformed ray. We derive the transformed ray's direction vector $\mathbf{d}_u$ and moment vector $\mathbf{m}_u$ as:

$$\mathbf{d}_u = (S^{-1}R^\top) \mathbf{d}, \quad \mathbf{m}_u = \mathbf{o}_u \times \mathbf{d}_u, \quad (\text{B.8})$$

where $\mathbf{o}_u = (S^{-1}R^\top) (\mathbf{o} - \boldsymbol{\mu})$ denotes the optical center in canonical space. Based on this, the square of $D_{\mu,\Sigma}(\mathbf{o}, \mathbf{d})$ can be efficiently calculated as:

$$(D_{\mu,\Sigma}(\mathbf{o}, \mathbf{d}))^2 = \frac{\mathbf{m}_u^\top \mathbf{m}_u}{\mathbf{d}_u^\top \mathbf{d}_u}. \quad (\text{B.9})$$

By substituting this distance square into Eq. [B.7](#) and then the derived transmittance value into Eq. [B.3](#) we compute the accumulated color along the ray.

B.3 RAY INTEGRAL OF ISOTROPIC 3D GAUSSIAN

Theorem 1. *Given a ray and a standard isotropic 3D Gaussian, the ray-Gaussian integral has a closed-form proportional to:*

$$\exp\left(-\frac{1}{2}D^2\right),$$

where D is the perpendicular distance from the Gaussian center to the ray.

Proof. We first apply a translation to the coordinate system such that the Gaussian $\mathcal{G}$ is centered at the origin, and the ray $\mathbf{r}_u$ is translated accordingly. Next, we rotate the entire scene so that the ray direction is aligned with the z -axis. Due to the translation-invariance of the integral and the isotropy of the standard Gaussian, these transformations do not affect the result of the integral.

The translated standard isotropic 3D Gaussian is defined as:

$$\mathcal{G}_{\mathbf{I},0}(\mathbf{u}) = \frac{1}{\rho} \exp\left(-\frac{1}{2}\mathbf{u}^\top \mathbf{u}\right), \quad (\text{B.10})$$

which matches Eq. [B.6](#). Let the z -axis aligned ray as:

$$\mathbf{r}_u(t) = \begin{bmatrix} \mathbf{o}_u \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad (\text{B.11})$$

the ray-Gaussian integral becomes:

$$\begin{aligned} \int_{t \in \mathbb{R}} \mathcal{G}_{\mathbf{I},0}(\mathbf{r}_u(t)) dt &= \frac{1}{\rho} \exp\left(-\frac{u_0^2 + v_0^2}{2}\right) \int_{t \in \mathbb{R}} \exp\left(-\frac{1}{2}t^2\right) dt \\ &= \frac{\sqrt{2\pi}}{\rho} \exp\left(-\frac{1}{2}D^2\right), \end{aligned} \quad (\text{B.12})$$

where $D^2 = \frac{u_0^2 + v_0^2}{2}$ denotes the squared perpendicular distance from the ray to the Gaussian center in canonical space. Thus, the integral is proportional to $\exp\left(-\frac{1}{2}D^2\right)$, as stated in theorem. $\square$

Note that the density function shared by our 3D primitives in the canonical space (see Eq. [B.6](#)) follows the same formulation as in the isotropic case. Given the world-coordinate ray $(\mathbf{o}, \mathbf{d})$ as the input of D , its value also depends on the world-space Gaussian parameters $\boldsymbol{\mu}, \Sigma$, which are used to transform the ray into the canonical coordinate system. Therefore, for each Gaussian contributing to the ray, we derive the transmittance as:

$$T = \frac{\sqrt{2\pi}\sigma}{\rho} \exp\left(-\frac{1}{2}(D_{\mu,\Sigma}(\mathbf{o}, \mathbf{d}))^2\right), \quad (\text{B.13})$$

which recovers Eq. [B.7](#) when picking $\rho = \sqrt{2\pi}$.

C APPENDIX: COOKBOOK OF GRADIENT COMPUTATION

C.1 CHAIN RULE

Recall that $T = \sigma\alpha$ is the ray-Gaussian transmittance (Eq. B.7), where σ is the opacity coefficient, and α is the ray-Gaussian integral. We denote $\kappa = D_{\mu, \Sigma}^2$ as the minimal squared Mahalanobis distance from the ray to the Gaussian distribution, S as the Gaussian scaling, and R as the rotation matrix in the world space. Additionally, given $\Sigma = RSS^\top R^\top$ as the 3D covariance matrix of the Gaussian in the world space, we have W_{pca} as the PCA whitening matrix satisfying:

$$W_{\text{pca}} = \Sigma^{-\frac{1}{2}} = S^{-1}R^\top, \quad \text{where } S^{-1} = \text{diag}\left(\frac{1}{s}\right). \quad (\text{C.1})$$

As illustrated in Fig. 2 we can apply the chain rule to find derivatives *w.r.t.* the Gaussian parameters, e.g, scaling vector s , rotation quaternion q , and mean μ :

$$\frac{\partial T}{\partial s} = \frac{\partial T}{\partial \kappa} \frac{\partial \kappa}{\partial W_{\text{pca}}} \frac{\partial W_{\text{pca}}}{\partial s}, \quad (\text{C.2})$$

$$\frac{\partial T}{\partial q} = \frac{\partial T}{\partial \kappa} \frac{\partial \kappa}{\partial W_{\text{pca}}} \frac{\partial W_{\text{pca}}}{\partial q}, \quad (\text{C.3})$$

$$\frac{\partial T}{\partial \mu} = \frac{\partial T}{\partial \kappa} \frac{\partial \kappa}{\partial \mu}. \quad (\text{C.4})$$

C.2 TWO-STAGE PROPAGATION

To reduce computational overhead in shared propagation, we design our gradient computation following the structured two-stage approach of 3DGS (Kerbl et al. 2023), comprising `RenderCUDA` (RC) and `PreprocessCUDA` (PC). The former performs ray-wise accumulation, efficiently accelerated by our Particle Bounding Frustum (PBF) association (see Section 2.2), while the latter operates with Gaussian-wise complexity, leveraging view frustum culling for additional speedup.

According to Eq. B.7 we can respectively write the gradient *w.r.t.* opacity coefficient and the shared partial derivative for Gaussian parameters in RC as:

$$\left. \frac{\partial T}{\partial \sigma} \right|_{\text{RC}} = \alpha, \quad \left. \frac{\partial T}{\partial \kappa} \right|_{\text{RC}} = -\frac{T}{2}. \quad (\text{C.5})$$

The remaining gradients to the Gaussian parameters can then be back-propagated from the squared distance, e.g., $\frac{\partial \kappa}{\partial s}, \frac{\partial \kappa}{\partial q}, \frac{\partial \kappa}{\partial \mu}$.

C.3 GRADIENTS FOR CANONICAL RAY

We then backpropagate gradients to the canonical ray parameters: the origin $\mathbf{o}_u$, direction $\mathbf{d}_u$, and moment $\mathbf{m}_u = \mathbf{o}_u \times \mathbf{d}_u$, as defined in Eq. B.8 in our paper. These quantities directly influence the Mahalanobis distance in Eq. B.9 and are critical for capturing canonical ray geometry.

$$\left. \frac{\partial \kappa}{\partial \mathbf{m}_u} \right|_{\text{RC}} = \left(\frac{2}{\mathbf{d}_u^\top \mathbf{d}_u} \right) \mathbf{m}_u, \quad (\text{C.6})$$

$$\left. \frac{\partial \kappa}{\partial \mathbf{o}_u} \right|_{\text{RC}} = \frac{\partial \kappa}{\partial \mathbf{m}_u} \times \mathbf{d}_u, \quad (\text{C.7})$$

$$\left. \frac{\partial \kappa}{\partial \mathbf{d}_u} \right|_{\text{RC}} = - \left(\frac{2\kappa}{\mathbf{d}_u^\top \mathbf{d}_u} \right) \mathbf{d}_u + \mathbf{o}_u \times \frac{\partial \kappa}{\partial \mathbf{m}_u}. \quad (\text{C.8})$$

C.4 GRADIENTS FOR WHITENING MATRIX

Recall that $\mathbf{d}_u = W_{\text{pca}} \mathbf{d}$ and $\mathbf{o}_u = W_{\text{pca}}(\mathbf{o} - \boldsymbol{\mu})$, where $\mathbf{o} = -R_c^\top \mathbf{t}_c$ denotes the world space optical center, and $\mathbf{d}$ denotes the ray direction. Once the gradients *w.r.t.* the ray parameters are obtained, we further propagate them to the PCA whitening matrix W_{pca} via outer products with the world-space vectors:

$$\begin{aligned} \text{RenderCUDA: } \left. \frac{\partial \kappa}{\partial W_{\text{pca}}} \right|_{\text{RC}} &= \frac{\partial \kappa}{\partial \mathbf{d}_u} \mathbf{d}^\top; \\ \text{PreprocessCUDA: } \left. \frac{\partial \kappa}{\partial W_{\text{pca}}} \right|_{\text{PC}} &= \frac{\partial \kappa}{\partial \mathbf{o}_u} (\mathbf{o} - \boldsymbol{\mu})^\top; \\ \text{Overall: } \frac{\partial \kappa}{\partial W_{\text{pca}}} &= \left. \frac{\partial \kappa}{\partial W_{\text{pca}}} \right|_{\text{RC}} + \left. \frac{\partial \kappa}{\partial W_{\text{pca}}} \right|_{\text{PC}}. \end{aligned} \quad (\text{C.9})$$

Here, the gradient backpropagated through $\mathbf{d}_u$ is accumulated ray-wise in `RenderCUDA`, while the gradient through $\mathbf{o}_u$ is computed Gaussian-wise in `PreprocessCUDA`. These two components are then summed to yield the complete gradient *w.r.t.* the PCA transformation matrix W_{pca} .

C.5 GRADIENTS FOR SCALING AND ROTATION

Thus, for scaling gradients $\frac{\partial \kappa}{\partial \mathbf{s}} = \frac{\partial \kappa}{\partial W_{\text{pca}}} \frac{\partial W_{\text{pca}}}{\partial \mathbf{s}}$, we have:

$$\left. \frac{\partial W_{\text{pca}}^{i,j}}{\partial s_k} \right|_{\text{PC}} = \begin{cases} -\frac{1}{s_k^2} R_{j,k} & \text{if } i = k \\ 0 & \text{otherwise} \end{cases}. \quad (\text{C.10})$$

To derive gradients for rotation $\frac{\partial \kappa}{\partial \mathbf{q}} = \frac{\partial \kappa}{\partial W_{\text{pca}}} \frac{\partial W_{\text{pca}}}{\partial \mathbf{q}}$, given the quaternion form $\mathbf{q} = q_r + q_i \mathbf{i} + q_j \mathbf{j} + q_k \mathbf{k}$, we can write the whitening matrix as:

$$W_{\text{pca}} = 2 \begin{pmatrix} \frac{1}{s_1} (\frac{1}{2} - q_j^2 - q_k^2) & \frac{1}{s_1} (q_i q_j + q_r q_k) & \frac{1}{s_1} (q_i q_k - q_r q_j) \\ \frac{1}{s_2} (q_i q_j - q_r q_k) & \frac{1}{s_2} (\frac{1}{2} - q_i^2 - q_k^2) & \frac{1}{s_2} (q_j q_k + q_r q_i) \\ \frac{1}{s_3} (q_i q_k + q_r q_j) & \frac{1}{s_3} (q_j q_k - q_r q_i) & \frac{1}{s_3} (\frac{1}{2} - q_i^2 - q_j^2) \end{pmatrix}. \quad (\text{C.11})$$

As a result, we derive the following end gradients for the quaternion $\mathbf{q}$ as:

$$\begin{aligned} \left. \frac{\partial W_{\text{pca}}}{\partial q_r} \right|_{\text{PC}} &= 2 \begin{pmatrix} 0 & \frac{q_k}{s_1} & -\frac{q_j}{s_1} \\ -\frac{q_k}{s_2} & 0 & \frac{q_i}{s_2} \\ \frac{q_j}{s_3} & -\frac{q_i}{s_3} & 0 \end{pmatrix}, \quad \left. \frac{\partial W_{\text{pca}}}{\partial q_i} \right|_{\text{PC}} = 2 \begin{pmatrix} 0 & \frac{q_j}{s_2} & \frac{q_k}{s_2} \\ \frac{q_k}{s_3} & -\frac{q_r}{s_3} & -\frac{2q_i}{s_3} \\ -\frac{2q_j}{s_3} & \frac{q_r}{s_3} & \frac{q_i}{s_3} \end{pmatrix}, \\ \left. \frac{\partial W_{\text{pca}}}{\partial q_j} \right|_{\text{PC}} &= 2 \begin{pmatrix} -\frac{2q_j}{s_1} & \frac{q_i}{s_1} & -\frac{q_r}{s_1} \\ \frac{q_i}{s_2} & 0 & \frac{q_k}{s_2} \\ \frac{q_r}{s_3} & \frac{q_k}{s_3} & -\frac{2q_i}{s_3} \end{pmatrix}, \quad \left. \frac{\partial W_{\text{pca}}}{\partial q_k} \right|_{\text{PC}} = 2 \begin{pmatrix} -\frac{2q_k}{s_1} & \frac{q_r}{s_1} & \frac{q_j}{s_1} \\ -\frac{q_r}{s_2} & -\frac{2q_k}{s_2} & \frac{q_j}{s_2} \\ \frac{q_i}{s_3} & \frac{q_j}{s_3} & 0 \end{pmatrix}. \end{aligned} \quad (\text{C.12})$$

C.6 GRADIENTS FOR 3D MEAN

Meanwhile, we compute end gradients *w.r.t.* the Gaussian 3D mean $\boldsymbol{\mu}$ as:

$$\left. \frac{\partial \kappa}{\partial \boldsymbol{\mu}} \right|_{\text{PC}} = -W_{\text{pca}}^\top \frac{\partial \kappa}{\partial \mathbf{o}_u}. \quad (\text{C.13})$$

C.7 NUMERICAL STABILITY OF OUR DIFFERENTIABLE FRAMEWORK

The choice of forward formulation, backward gradient computation, and intermediate cache variables critically affects numerical stability. Prior approaches such as GOF (Yu et al., 2024b) suffer from instability in backward propagation and therefore rely on additional 3D filtering heuristics to stabilize training. In contrast, our closed-form formulation, together with its exact gradient chain, avoids such instability altogether, eliminating the need for any auxiliary filtering.



Figure C.1: **Artifacts without 3D filtering in GOF.** (Left) when degenerated Gaussians are not filtered in GOF, and (Right) the stable results obtained when applying our forward and backward formulations.

The key distinction from our gradient and rendering pipeline lies in that prior methods adopt an alternative expansion of $\mathbf{m}_u^\top \mathbf{m}_u$. Ideally, disregarding numerical issues, the following two forms are geometrically equivalent:

$$\mathbf{m}_u^\top \mathbf{m}_u = \|\mathbf{o}_u\|^2 \|\mathbf{d}_u\|^2 - (\mathbf{o}_u^\top \mathbf{d}_u)^2. \quad (\text{C.14})$$

However, when a 3D Gaussian has one axis scaled close to zero, the canonical-space vectors $\mathbf{o}_u$ and $\mathbf{d}_u$ become nearly parallel. In this case, $\cos(\angle(\mathbf{o}_u, \mathbf{d}_u)) \rightarrow 1$, while both $\|\mathbf{o}_u\|$ and $\|\mathbf{d}_u\|$ diverge. This makes the term $\|\mathbf{o}_u\|^2 \|\mathbf{d}_u\|^2 - (\mathbf{o}_u^\top \mathbf{d}_u)^2$ highly prone to numerical overflow and, in practice, can even yield negative values and further lead to instable backward propagation as well.

Fig. C.1 (Left) demonstrates the numerical artifacts produced by GOF. When it does not apply additional 3D filtering to remove degenerated Gaussians, severe artifacts appear. The snowflake noise arises from numerical overflow in Eq. C.14 where negative values are prematurely returned and extremely small positive values lead to amplified Gaussian intensity. Since the EWA approximation with Jacobian and conic remains numerically stable, the artifacts mainly stem from inaccurate maximum-response computation, which also induces gridline patterns.

In contrast, our CUDA rasterizer implementation computes both the cross product and its gradients directly (Eq. B.9 (forward) and Eqs. C.6-C.9 (backward)), thereby fundamentally avoiding this numerical issue. As a result, the forward pass is artifact-free, while the backward pass yields accurate gradients. Fig. C.1 (Right) illustrates the results of applying our forward and backward formulations to GOF: even without additional 3D filtering to exclude degenerated Gaussians (i.e., flat disk-like or needle Gaussians), our approach effectively eliminates the artifacts caused by numerical instability.

D APPENDIX: ADDITIONAL DETAILS IN ASSOCIATION

Section D.1 provides our detailed derivation. Section D.2 provides implementation considerations required for stable PBF under large-FoV and omnidirectional cameras. Section D.3 clarifies the derivational relationships and explicitly states the conditions under which prior formulas coincide with ours.

D.1 PARTICLE BOUNDING FRUSTUM (PBF) SOLVER

Recall Eq. 8 we define $P = (VH)^\top \in \mathbb{R}^{4 \times 4}$ as the plane transformation matrix from the view space to the canonical space. Take the unknown bounds for $c = \tan \theta_{1,2}$ as example, we have:

$$\mathbf{g}_u = -P_0 + cP_2 = \begin{bmatrix} -P_{00} + cP_{02} \\ -P_{10} + cP_{12} \\ -P_{20} + cP_{22} \\ -P_{30} + cP_{32} \end{bmatrix}, \quad (\text{D.1})$$

where P_i denotes i -th row of the matrix, expressed as a 4×1 column vector, and P_{ij} further denotes j -th element. Recall Eq. 9 assuming we are computing the bounding box of the 1-standard-deviation contour ellipsoid of the Gaussian, i.e., $\lambda = 1$, then we have the following quadratic equation:

$$((\mathbf{f} \circ P_2^\top) \cdot P_2) c^2 - 2((\mathbf{f} \circ P_0^\top) \cdot P_2) c + (\mathbf{f} \circ P_0^\top) \cdot P_0 = 0, \quad (\text{D.2})$$

where $\mathbf{f} = (1, 1, 1, -1)$ is a 1×4 row vector. Here we denote the equation as:

$$\mathcal{T}_{22}c^2 - 2\mathcal{T}_{02}c + \mathcal{T}_{00} = 0, \quad (\text{D.3})$$

where $\mathcal{T}$ is a 3×3 upper triangular matrix defined with $\mathcal{T}_{ij} = (\mathbf{f} \circ \mathbf{P}_i^\top) \cdot \mathbf{P}_j$.

Similarly, for the bounds of ϕ , we have $c = \tan \phi_{1,2}$ satisfying:

$$\mathcal{T}_{22}c^2 - 2\mathcal{T}_{12}c + \mathcal{T}_{11} = 0. \quad (\text{D.4})$$

Interestingly, given a camera-view configuration $[R_c \mid \mathbf{t}_c]$, we find that the parameter matrix $\mathcal{T}$ can be efficiently computed from the Gaussian's view-space covariance $\Sigma_c = R_c R S S^\top R^\top R_c^\top$ and 3D mean $\mu_c = R_c \mu + \mathbf{t}_c$:

$$\mathcal{T}_{ij} = \lambda^2 \Sigma_c^{i,j} - (\mu_c \mu_c^\top)^{i,j}. \quad (\text{D.5})$$

Thus, we can compute the angular bounds (defined by their tangent-space center and extent) from:

$$\begin{aligned} \left(\frac{\tan \theta_1 + \tan \theta_2}{2}, \frac{\tan \phi_1 + \tan \phi_2}{2} \right) &= \left(\frac{\mathcal{T}_{02}}{\mathcal{T}_{22}}, \frac{\mathcal{T}_{12}}{\mathcal{T}_{22}} \right), \\ \|\tan \theta_1 - \tan \theta_2\| &= \frac{2\sqrt{\mathcal{T}_{02}^2 - \mathcal{T}_{22}\mathcal{T}_{00}}}{|\mathcal{T}_{22}|}, \\ \|\tan \phi_1 - \tan \phi_2\| &= \frac{2\sqrt{\mathcal{T}_{12}^2 - \mathcal{T}_{22}\mathcal{T}_{11}}}{|\mathcal{T}_{22}|}. \end{aligned} \quad (\text{D.6})$$

If the discriminant of either quadratic is negative—indicating that the camera's optical center lies inside the Gaussian ellipsoid and the angular bounds are undefined—we conservatively clamp the frustum angles using the camera's maximum field of view (FoV).

Note that this result is directly computed from the 3D view-space true covariance, in contrast to methods like EWA and UT, which rely on intermediate conic approximations in screen space. Our PBF formulation offers both exactness and efficiency.

D.2 DETAILS IN COMPUTING CSF-PBF INTERSECTION

We uniformly sample spherical angles within the maximum field of view (FoV). The resulting x , y , and z coordinates—i.e., the ray direction vectors $\mathbf{d}$ —are projected onto a unit sphere and then transformed into pixel space for inverse color interpolation (see Sec. E). These sampled rays are then grouped according to their corresponding CSFs for subsequent Gaussian association.

In most scenes using a pinhole camera with a limited FoV, the angular bounds in PBF typically lie within $[-\frac{\pi}{2}, \frac{\pi}{2}]$. In such cases, since the $\tan$ function is monotonically increasing over this interval, the intersections between PBF and CSFs can be computed by directly comparing the $\tan$ values of their respective angular bounds. However, in wide FoV scenarios, even when the incidence angle formed by the Gaussian centers remains within $[-\frac{\pi}{2}, \frac{\pi}{2}]$, the angular bounds used in PBF may still fall outside the monotonic region of the $\tan$ function. In these cases, directly comparing $\tan$ values may yield incorrect PBF-CSF intersections, such as mistakenly matching angular bounds as $[\theta'_2, \theta_1]$ (see Fig. D.1 for an example).

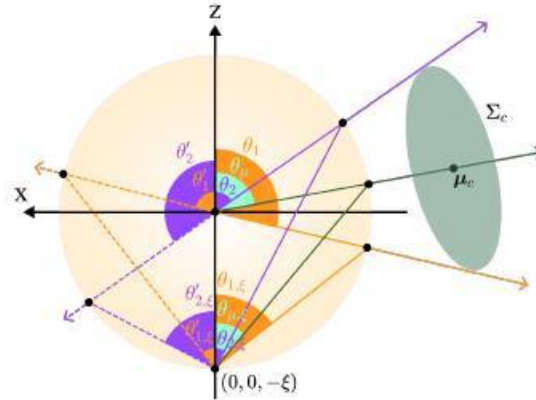


Figure D.1: Mirror Transformation of PBF Angular Bounds. Four candidate mirrored angles $\theta_{1,\xi}, \theta'_{1,\xi}, \theta_{2,\xi}, \theta'_{2,\xi}$ are generated from a given pair of bounds. Given a Gaussian centered at μ_c , the two closest to the corresponding Gaussian's mirrored angle $\theta_{\mu,\xi}$ are used as PBF bounds for robust intersection with mirror-transformed CSF partitions.

To resolve the correct intersections, we introduce a mirror transformation applied to the tangent values of both PBF angular bounds and CSF partitions. Inspired by the CMEI camera model (Mei & Rives 2007), we shift the optical center using a mirror parameter ξ , leading to the following transformation:

$$\tan \theta_\xi = \frac{\tan \theta}{1 + \xi \frac{z}{\|z\|} \sqrt{1 + \tan^2 \theta}} = \frac{\sin \theta}{\cos \theta + \xi}, \quad (\text{D.7})$$

where θ_ξ represents the mirror-transformed angle of θ on the xz -plane, using a mirror parameter ξ . We set $\xi = 1$ throughout all experiments.

Given a pair of tangent value $[\tan \theta_1, \tan \theta_2]$ from our PBF solver (See Equation. D.6), this transformation yields four candidate mirror angles, as illustrated in Fig. D.1. The Gaussian center, in contrast, corresponds to a unique mirror-transformed angle $\theta_{\mu, \xi}$. We exploit this property by selecting the two candidate angles closest to $\theta_{\mu, \xi}$. These two then define the effective lower and upper bounds for PBF, i.e., $[\theta_{1, \xi}, \theta_{2, \xi}]$, which are subsequently compared with CSF partitions to compute the correct intersections.

This trick ensures robust and consistent angular comparisons, even beyond the monotonic domain of the $\tan$ function. As reported in our timing analysis (Tab. K.5), the pre-processing pipeline—including the mirror-angle-based PBF-CSF intersection computation and the canonical transformation for all Gaussians—takes less than 0.13 ms in total. This confirms the efficiency of our approach, even when applied to thousands of Gaussians across wide FoV scenarios.

D.3 RELATION TO PRIOR WORKS

Substituting the perspective screen space coordinates:

$$u = \tan \theta, \quad v = \tan \phi$$

into our formulation yields expressions identical to those in 2DGS (Huang et al. 2024a) and HTGS (Hahlbohm et al. 2025). This is expected as a perspective alternative of our angular-domain formulation.

Concretely, 2DGS and HTGS start from a screen-space or tangent-plane parameterization and therefore rely on assumptions that hold only for perspective (pinhole) cameras, e.g., geometric identities that arise when rays are implicitly tied to a planar, z -parallel image plane. This assumption holds only for perspective cameras and does not extend to fisheye or omnidirectional systems, where the corresponding surface in screen space is curved. In contrast, our derivation begins from a camera-agnostic angular parameterization in 3D and directly parameterizes Camera Sub-Frustums (CSFs) and Particle Bounding Frustums (PBFs). These different starting assumptions change the known quantities and constraints available during derivation, and thus the mathematical conditions under which closed-form bounds and culling guarantees remain valid for generic cameras.

We’d like to emphasize that mathematics in itself is a general tool, but deriving these bounds under weaker assumptions (i.e., without relying on screen-space structure) and validating them with comprehensive experiments across both pinhole and wide-FOV camera families constitute an essential part of our contribution.

This difference in parameterization also affects the downstream computational pipeline. Gaussian association, tiling/duplication, and sorting depend on how Gaussians map to tiles or bins. Using an angular (CSF-based) domain yields a different bucketing structure compared to pinhole screen-space tiling. This is practically important for large-FoV scenes: under generic cameras a Gaussian’s bound may appear partially “behind” the camera in screen coordinates (see Fig. D.1), causing screen-space methods to produce mirrored or incorrect tangent-based bounds (dashed line in Fig. D.1). To robustly handle these cases, we introduce mirror parameters in the association step (Section D.2) and incorporate the Gaussian center into sorting and correction—components unnecessary in perspective-only derivations.

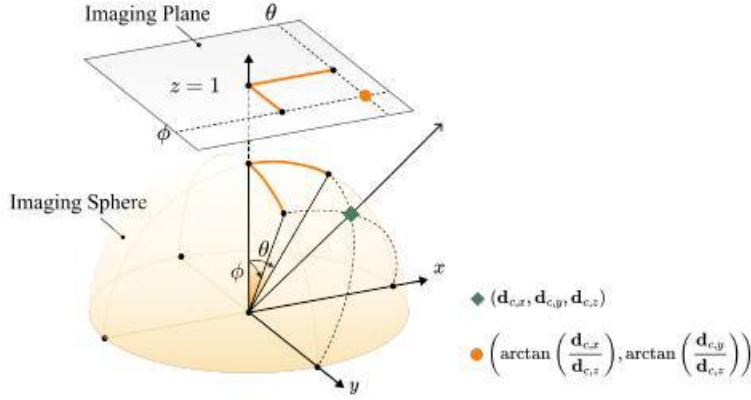


Figure E.1: **Spherical Angles Defining PBF and BEAP.** A view-space direction $\mathbf{d}_c$ is first normalized onto the optical sphere (green spade), followed by angular projections onto the xz and yz planes. The orange point denotes the resulting coordinate (θ, ϕ) on the unfolded BEAP imaging plane.

E APPENDIX: BEAP IMAGE INVERSE INTERPOLATION

To prepare training data, we sample rays to comprehensively cover the original image FoVs and use them as input to our CUDA renderer. Simultaneously, we interpolate the ground truth color for each input ray to ensure accurate and uniform supervision during training.

The projection of sampled rays onto observed images begins with transforming the incidence angle projections θ and ϕ into normalized ray direction vectors $\mathbf{d} = (x, y, z)^\top$, expressed as:

$$x = \frac{\sin \theta \cos \phi}{\sqrt{\sin^2 \theta \cos^2 \phi + \cos^2 \theta \sin^2 \phi + \cos^2 \theta \cos^2 \phi}}, \quad (\text{E.1})$$

$$y = \frac{\cos \theta \sin \phi}{\sqrt{\sin^2 \theta \cos^2 \phi + \cos^2 \theta \sin^2 \phi + \cos^2 \theta \cos^2 \phi}}, \quad (\text{E.2})$$

$$z = \frac{\cos \theta \cos \phi}{\sqrt{\sin^2 \theta \cos^2 \phi + \cos^2 \theta \sin^2 \phi + \cos^2 \theta \cos^2 \phi}}. \quad (\text{E.3})$$

Several camera models are provided to subsequently project the normalized rays $(x, y, z)^\top$ into pixel space $(x_p, y_p, 1)^\top$.

Pinhole Camera Model. For pinhole camera models, we first have $\frac{x}{z} = \tan \theta$ and $\frac{y}{z} = \tan \phi$. Thus, we have:

$$\begin{bmatrix} x_p \\ y_p \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \tan \theta \\ \tan \phi \\ 1 \end{bmatrix}. \quad (\text{E.4})$$

KB Fisheye Model. We have the incidence angle φ derived from its projections:

$$r = \sqrt{\tan^2 \theta + \tan^2 \phi} \quad \text{and} \quad \varphi = \arctan(r). \quad (\text{E.5})$$

And we have the fisheye distorted angle:

$$\varphi_d = \varphi(1 + k_1 \varphi^2 + k_2 \varphi^4 + k_3 \varphi^6 + k_4 \varphi^8). \quad (\text{E.6})$$

As a result, the rays are projected onto pixel space, allowing for inverse color interpolation from:

$$\begin{bmatrix} x_p \\ y_p \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{f_x \varphi_d}{r} & 0 & c_x \\ 0 & \frac{f_y \varphi_d}{r} & c_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \tan \theta \\ \tan \phi \\ 1 \end{bmatrix}. \quad (\text{E.7})$$

F APPENDIX: HIGH-LEVEL ALGORITHMIC SUMMARY

We provide a high-level overview of our volumetric rendering pipeline (See Alg. 1 and Alg. 2). For clarity, we follow the structure of 3DGS (Kerbl et al., 2023) to illustrate differences: newly introduced modules or replaced ones are highlighted in green, and significantly modified ones in yellow.

ALGORITHM 1: GPU Software Asso. & Render.

```

Function 3DGEER ( $\mu, \mathbf{s}, \mathbf{q}, c, \sigma, \mathbf{r}_c, V$ ):
  CullGaussian( $\mu, V$ )                                ▷ Frustum Culling
   $F \leftarrow \text{InitCSFs}(w, h, K)$                       ▷ Partition
   $\Sigma_c, \mu_c \leftarrow \text{ViewspaceGaussians}(\mathbf{s}, \mathbf{q}, V)$   ▷ Transform
   $F_{ID} \leftarrow \text{PBF\_Intersect}(\Sigma_c, \mu, F)$         ▷ Sub-frustum Asso.
   $W_{pca} \leftarrow \text{PCA\_Whitening}(\mathbf{s}, \mathbf{q})$               ▷ Transform
   $\mathcal{G}_{ID}, F_K \leftarrow \text{DuplicateWithKeys}(W_{pca}, F_{ID})$ 
  ▷ Indices (Gaussian) & Keys (FrustumID+Depth)
  SortByKeys( $\mathcal{G}_{ID}, F_K$ )                                ▷ Globally Depth Sort
   $R \leftarrow \text{IdentifyFrustumRanges}(F, F_K)$ 
   $I \leftarrow 0$                                           ▷ Init Canvas (w/ Rays)
  for all CSFs  $f \leftarrow I$  do
    for all Rays  $i \leftarrow f$  do
       $r \leftarrow \text{GetFrustumRange}(R, f)$ 
       $\mathbf{m}_u, \mathbf{d}_u \leftarrow \text{Cano\_Ray}(W_{pca}, \mu, i)$ 
       $D_{\mu, \Sigma} \leftarrow \text{Dist}(\mathbf{m}_u, \mathbf{d}_u)$               ▷ Transform
       $I[i] \leftarrow \text{BlendInOrder}(i, \mathcal{G}_{ID}, r, F_K, D_{\mu, \Sigma}, c, \sigma)$   ▷ Mah-distance
    end
  end
  return  $I$ 
  ▷ Alpha Blending

```

ALGORITHM 2: 3DGEER Initialization and Training Framework

w, h : width and height of the training images

K, V : intrinsics and extrinsic meta info.

```

 $\mu \leftarrow \text{SfM Points}$                                 ▷ Init Gaussian Attributes
 $\mathbf{s}, \mathbf{q} \leftarrow \text{InitAttributes}()$                     ▷ Position
 $c, \sigma \leftarrow \text{InitCoefficients}()$                 ▷ Scaling, Rotation
 $\text{while not converged do}$                                 ▷ Color, Opacity
   $\mathbf{r}_c, \hat{C} \leftarrow \text{BEAP}(w, h, K)$                     ▷ View Rays / Colors
   $C \leftarrow 3DGEER(\mu, \mathbf{s}, \mathbf{q}, c, \sigma, \mathbf{r}_c, V)$         ▷ Asso. & Render
   $\mathcal{L} \leftarrow \text{Loss}(C, \hat{C})$                             ▷ See Alg. 1
   $\mu, \mathbf{s}, \mathbf{q}, c, \sigma \leftarrow \text{Adam}(\nabla \mathcal{L})$         ▷ Loss
  if IsRefinementIteration( $i$ ) then                    ▷ Backprop & Step
     $\mu, \mathbf{s}, \mathbf{q}, c, \sigma \leftarrow \text{Optimize}(\mu, \mathbf{s}, \mathbf{q}, c, \sigma)$   ▷ Kerbl et al. (2023)
  end
end

```

G APPENDIX: ADDITIONAL IMPLEMENTATION DETAILS

Baselines. We compare 3DGEER with leading volumetric particle rendering methods targeting projective approximation errors in 3DGS (Kerbl et al., 2023). For large FoV fisheye inputs, we evaluate against FisheyeGS (Liao et al., 2024), EVER (Mai et al., 2025), and 3DGUT (Wu et al., 2025). Under pinhole settings, we additionally compare with 3DGS and 3DGRT (Moënné-Loccoz et al., 2024).

Implementation Details. To ensure fair comparison, we align our implementation with the original 3DGS in terms of initialization and hyperparameters on the ScanNet++ and MipNeRF360 datasets. For ZipNeRF, we adopt the training configurations of EVER and SMERF (Duckworth et al. 2024) for all candidate methods. Similar to most ray-based frameworks, our method does not directly access gradients of `mean2D` to control densification or pruning. Instead, we optimize using gradients of the view-space `mean3D`. Unless otherwise noted, all models are trained with consistent supervision strategies and architectural settings. For fisheye datasets (ScanNet++ and ZipNeRF), we precompute bijective grid mappings (Guo et al. 2025) and reproject all rendered outputs to the native KB-Fisheye image space (Kannala & Brandt 2006) before evaluation. Experiments on ScanNet++ and MipNeRF360 are conducted using full-resolution images. For ZipNeRF, due to the high frame count and extremely large fisheye resolution, we use the maximum resolution (see Tab. 3) that allows all baselines to run reliably on a local desktop setup.

Central and peripheral regions in ScanNet++ evaluation. The rays sampled in the central part follows the range:

$$\theta \in [-\frac{w}{2f_x}, \frac{w}{2f_x}], \phi \in [-\frac{h}{2f_y}, \frac{h}{2f_y}].$$

And the Peripheral part takes the rays sampled in remaining angle ranges. Here w, h indicate the width and height of the image and f_x, f_y are the focal lengths defined in the original KB camera model.

Cross-camera generalization evaluation on ZipNeRF. ZipNeRF provides two sets of data: fisheye raw images and their corresponding undistorted frames, which are generated by applying distortion correction and cropping the central regions. Although the frames are aligned on a per-frame basis, camera poses and point clouds were independently reconstructed using COLMAP (Schönbberger & Frahm 2016) for each set. Due to COLMAP’s global optimization independently refining camera trajectories, applying a single rigid transformation to align the two pose sets is insufficient and may introduce alignment errors. To ensure reliable cross-camera evaluation without introducing errors from structure-from-motion (SfM), we always use the same set of test-time camera poses as those used during training. In cross-camera experiments, we vary only the intrinsic parameters, while keeping the extrinsic poses fixed. This ensures fair comparisons without entangling pose inconsistencies arising from preprocessing.

H APPENDIX: EWA SPLATTING PRELIMINARIES

From camera coordinates to ray coordinates. As the coordinate definitions in EWA Splatting (Zwicker et al. 2001), we have the following mapping function from the camera (view) space points $\mathbf{t}$ to the ray space points $\mathbf{x}$ for projective transformation in pinhole camera models, *i.e.*, $\tau: \mathbf{t} \rightarrow \mathbf{x}$:

$$\mathbf{x} = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \end{bmatrix} = \tau(\mathbf{t}) = \begin{bmatrix} t_0/t_2 \\ t_1/t_2 \\ \|(t_0, t_1, t_2)^\top\| \end{bmatrix}, \quad (\text{H.1})$$

$$\mathbf{t} = \begin{bmatrix} t_0 \\ t_1 \\ t_2 \end{bmatrix} = \tau^{-1}(\mathbf{x}) = \begin{bmatrix} x_0/l \cdot x_2 \\ x_1/l \cdot x_2 \\ 1/l \cdot x_2 \end{bmatrix}, \quad (\text{H.2})$$

$$, \text{ where } l = \|(x_0, x_1, 1)^\top\|. \quad (\text{H.3})$$

These mappings are not affine, which requires computationally expensive (Westover, 1990; Mao et al. 1995) or locally approximated solutions (Zwicker et al. 2001; Kerbl et al. 2023) to calculate integral in the ray space.

Local affine approximation. EWA splatting by (Zwicker et al. 2001) introduced local affine approximation, denoted as τ_k , to keep the linear effects and the k -indexed reconstruction kernels as Gaussians in 2D screen space. It is defined by the first two terms of the Taylor expansion of τ at the camera space point $\mathbf{t}_k$, then the ray space point $\mathbf{x}$ near around $\mathbf{x}_k$ can be estimated by:

$$\mathbf{x} = \tau(\mathbf{t}) = \mathbf{x}_k + \mathbf{J}_\tau^{\mathbf{t}_k}(\mathbf{t} - \mathbf{t}_k), \quad (\text{H.4})$$



Figure I.1: **Qualitative Results on the Stump (Mip-NeRF360).** 3DGEER outperforms both 3DGS-based (Gaussian Splatting) and 3DPRT-based (Particle Ray Tracing) methods on novel view synthesis benchmarks under (Left) MipNeRF360 (pinhole), while maintaining high runtime efficiency. Larger differences are visible in the colored boxes.

where $\mathbf{J}_\tau^{\mathbf{t}_k}$ is the Jacobian matrix given by the partial derivatives of τ at the point $\mathbf{t}_k$. For the k -indexed 3D Gaussian $\mathcal{G}_{\Sigma, \mu}$ in a certain camera coordinate, using EWA local affine transformation, the approximated density function can be expressed as a 3D Gaussian $\mathcal{G}_{\Sigma_k, \mathbf{x}_k}$ in the ray coordinate. For $\forall \mathbf{x}$ around $\mathbf{x}_k$, we have:

$$\mathcal{G}_{\Sigma_k, \mathbf{x}_k}(\mathbf{x}) = \frac{1}{(2\pi)^{\frac{3}{2}} |\Sigma_k|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mathbf{x}_k)^\top \Sigma_k^{-1}(\mathbf{x} - \mathbf{x}_k)\right), \quad (\text{H.5})$$

$$\text{where } \Sigma_k = \mathbf{J}_\tau^{\mathbf{t}_k} \Sigma (\mathbf{J}_\tau^{\mathbf{t}_k})^\top. \quad (\text{H.6})$$

Integrating a 3D Gaussian $\mathcal{G}_{\Sigma_k, \mathbf{x}_k}^{(3)}$ along one coordinate axis yields a 2D Gaussian $\mathcal{G}_{\Sigma_k, \hat{\mathbf{x}}_k}^{(2)}$, hence:

$$\int_{\mathbb{R}} \mathcal{G}_{\Sigma_k, \mathbf{x}_k}^{(3)}(\mathbf{x}) dx_2 = \mathcal{G}_{\Sigma_k, \hat{\mathbf{x}}_k}^{(2)}(\hat{\mathbf{x}}), \quad (\text{H.7})$$

where $\hat{\mathbf{x}} = \begin{bmatrix} x_0 \\ x_1 \end{bmatrix}$, and the mean $\hat{\mathbf{x}}_k = \begin{bmatrix} x_{k,0} \\ x_{k,1} \end{bmatrix}$, and the 2×2 variance matrix $\hat{\Sigma}_k$ is easily obtained from the 3×3 matrix Σ_k by skipping the third row and column.

Splatting approximation error. Although we can not expect the density function in ray space still obey the Gaussian distribution when we apply general projective transformation to the original view space, exact density can be derived through substitution via Eq. [H.2](#)

$$\begin{aligned} \mathcal{G}_{\Sigma, \mu}(\mathbf{t}) &= \mathcal{G}_{\Sigma, \mu}(\tau^{-1}(\mathbf{x})) \\ &= \frac{1}{(2\pi)^{\frac{3}{2}} |\Sigma_k|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(\tau^{-1}(\mathbf{x}) - \mathbf{t}_k)^\top \Sigma_k^{-1}(\tau^{-1}(\mathbf{x}) - \mathbf{t}_k)\right), \end{aligned} \quad (\text{H.8})$$

which means that the approximation error can be estimated through Monte Carlo integration of the following equation:

$$\mathcal{E}_k = \int_{\mathbb{R}} |\mathcal{G}_{\Sigma, \mu}(\tau^{-1}(\mathbf{x})) - \mathcal{G}_{\Sigma_k, \mathbf{x}_k}(\mathbf{x})| dx_2. \quad (\text{H.9})$$

Note that x_2 represents the dimension along the ray direction in the ray coordinate.

I APPENDIX: ADVANTAGE OVER PRIOR RAY-TRACING METHODS

Pure ray-tracing (e.g., 3DGRT) can be ideally projective-exact, but in practice the BVH adopted in 3DGRT for intersection tests uses inscribed proxy icosahedron for faster speed, which is tight at vertices but face centers lie $\sim 20\%$ inward based on icosahedron's property. Proxy shapes may degrade intersection accuracy and further give worse accumulated gradients during density control and eventually lead to incorrect geometry (e.g., missing leaves in Fig. [I.1](#)). Further, intersection tests for far, tiny, and cluttered Gaussians are limited by floating-point precision that causes noisy rendering. (E.g., in Fig. 15 of the 3DGRT paper, native rendering in the leftmost block contains visible speckles in the tree and background areas.) Even 3DGUT with UT approximation outperforms 3DGRT on



Figure J.1: **Qualitative Results on the Aria Dataset.** Our method produces sharper floor textures in indoor scenes and clearer distant tree details in outdoor scenes.

Table J.1: **Quantitative comparison on Aria.** Higher PSNR/SSIM and lower LPIPS are better.

Method	Scene		PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
3DGEER (Ours)	livingroom	(indoor)	36.398	0.933	0.226
	steakhouse	(outdoor)	30.662	0.881	0.246
FisheyeGS	livingroom	(indoor)	35.686	0.927	0.231
	steakhouse	(outdoor)	29.989	0.869	0.261

MipNeRF360. For validation, we use 3DGRT’s ray-tracing and rendering engine and our 3DGEER rendering pipeline to render from the same pretrained MipNeRF360-stump and compare on PSNR $\uparrow$ /SSIM $\uparrow$ /LPIPS $\downarrow$: 3DGRT- 25.79/ 0.745/ 0.264; ours- 26.83/ 0.774/ 0.246 (+1.04/ +0.029/ -0.018). 3DGEER builds a projective-exact and efficient pipeline and further optimizes with BEAP color supervision, achieving much faster speed and better accuracy (Tab. 4).

J APPENDIX: RESULTS ON EGOCENTRIC DATASET

Recent egocentric devices typically feature wide FoVs, but existing pipelines struggle to maintain reconstruction quality when using full-resolution images or when restricted to cropped views. The challenge is further compounded by low-light imaging conditions—where limited light intake, especially with small sensors and aperture-constrained lenses, exacerbates degradation.

We evaluate on the Aria egocentric dataset ($\sim 110^\circ$ FoV, circular shape). Our method achieves an average PSNR of 33.5dB, consistently surpassing FisheyeGS (32.8dB) across all metrics (Tab. J.1). Qualitatively, our reconstructions preserve sharper details in peripheral regions for the indoor scene (Fig. J.1 Left) and recover distant structures such as trees in the outdoor scene (Fig. J.1 Right). The performance gain (+0.7 dB) lies between the improvements observed in pinhole datasets (e.g., MipNeRF360: +0.3dB; ZipNeRF-cross: +2.0dB) and in highly distorted wide-FoV datasets (e.g., ScanNet++/ZipNeRF 180° diagonal FoV: +0.9–4.8dB), which aligns with Aria’s moderate FoV and distortion level.

K APPENDIX: ADDITIONAL MATERIAL AND RESULTS

K.1 ADDITIONAL MATERIAL.

Code, video, and an HTML page are provided in the Supplementary Material.

K.2 ADDITIONAL RESULTS.



Figure K.1: **Out-of-Distribution (OOD) Region in Fig. 6** 3DGS collapses to black due to the numerical issue of $\tan 90^\circ \rightarrow \infty$, while our method robustly renders views up to 180° , successfully to reconstruct the green chair, where splat-based methods (e.g., FisheyeGS) fail. Similarly, in Fig. 6 (zoom-in), our method also preserves structures like the corridor and fridge, table and door, or window and cabinet.



Figure K.2: **Extreme-FoV Generalization.** (Left) Comparison between our method and 3DGS under extreme-FoV pinhole (170°) views (trained on undistorted central regions). (Right) Zoom-out results of Fig. 6 showing the complete circular 180° FoV, where our method generalizes robustly to out-of-distribution regions.

Table K.1: **MipNeRF360 Quantitative Results by Sequence.** The overall results are shown in Tab. 4

Method	Metric	Bicycle	Bonsai	Counter	Garden	Kitchen	Stump	Flowers	Room	Treehill
Ours	PSNR↑	25.63	32.23	29.05	27.60	31.78	26.83	21.83	32.09	22.83
	SSIM↑	0.774	0.946	0.910	0.860	0.931	0.774	0.613	0.946	0.637
	LPIPS↓	0.223	0.175	0.195	0.136	0.120	0.246	0.340	0.108	0.344

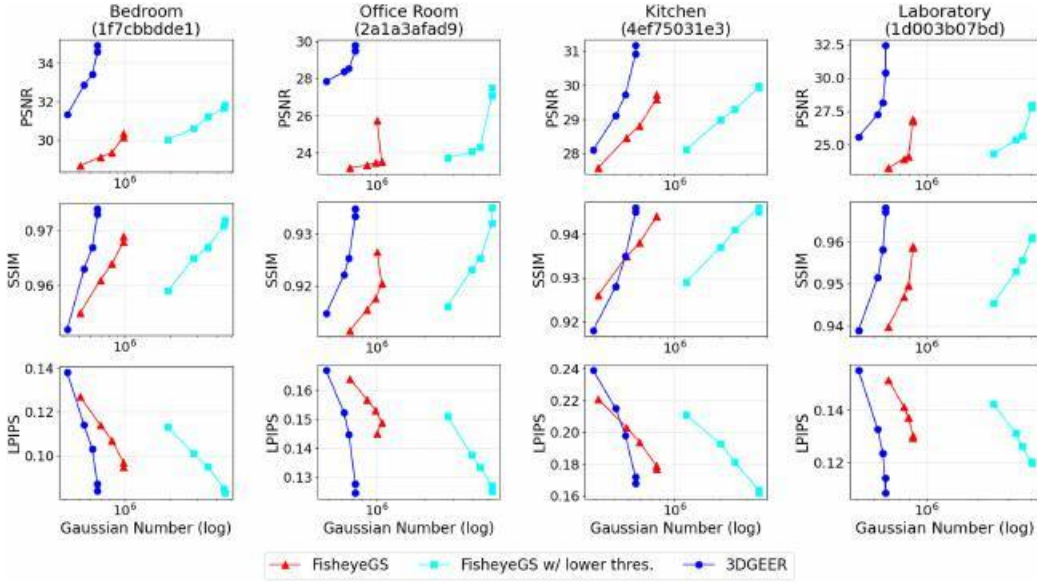


Figure K.3: **PSNR $\uparrow$ SSIM $\uparrow$ LPIPS $\downarrow$ Trends When Scaling-Up Gaussian Counts.** Brute-force scaling fails to close the gap in projective exactness.

Table K.2: **ScanNet++ Quantitative Results by Sequence (Full FoV).** The overall results are shown in Tab. 2.

Scene	EVER			FisheyeGS			3DGUT			3DGEER (Ours)		
	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS
Storage	28.24	0.916	0.164	24.74	0.932	0.146	29.26	0.934	0.147	29.18	0.940	0.141
Office Room	27.80	0.899	0.169	25.75	0.927	0.145	29.37	0.928	0.145	29.78	0.935	0.125
Bedroom	32.60	0.948	0.131	31.95	0.970	0.095	34.65	0.967	0.112	34.94	0.974	0.084
Kitchen	29.77	0.924	0.213	29.72	0.944	0.177	30.78	0.941	0.196	31.17	0.946	0.168
Laboratory	28.96	0.936	0.156	26.91	0.959	0.129	29.12	0.948	0.151	32.43	0.968	0.114

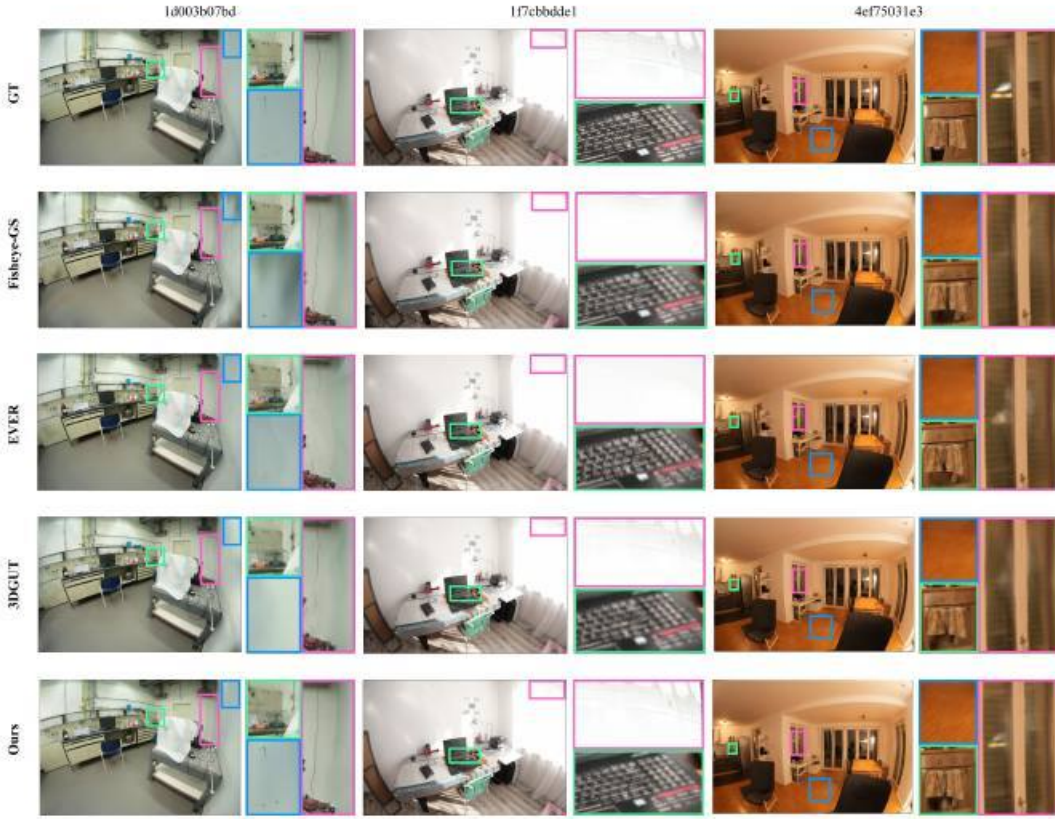


Figure K.4: **Qualitative Results on the ScanNet++ Dataset.** Larger differences are visible in the zoomed-in colored boxes, particularly in structural details and artifact regions. Our method delivers superior peripheral reconstruction compared to FisheyeGS, exhibits fewer artifacts and sharper details than EVER, and produces crisper object boundaries and textures—such as wires, keyboards, and wooden floors—when compared to 3DGUT.

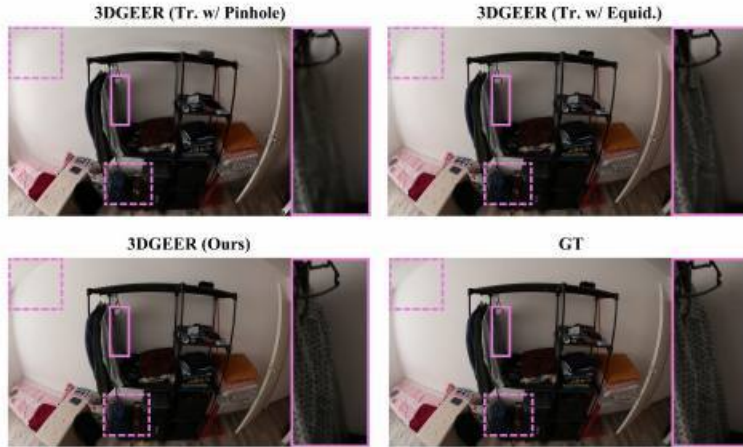


Figure K.5: **Impact of BEAP supervision.** Visual comparison on ScanNet++ shows that BEAP’s balanced angular sampling produces smoother gradients and better preservation of geometric and texture details than pinhole or equidistant sampling. (Tab. K.8 shows quantitative full stats).

Table K.3: **Quantitative Results on the ScanNet++**. PSNR $\uparrow$ SSIM $\uparrow$ LPIPS $\downarrow$ Trends When Scaling-Up Gaussian Counts. Bedroom, Office Room, Kitchen, and Laboratory in order.

Iter	FisheyeGS				FisheyeGS w/ Low Thres.				3DGEER			
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Gau $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Gau $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Gau $\downarrow$
4k	28.67	0.955	0.127	512K	30.03	0.959	0.113	1914K	31.33	0.952	0.138	422K
7k	29.12	0.961	0.114	695K	30.61	0.965	0.101	2829K	32.87	0.963	0.114	543K
10k	29.36	0.964	0.107	827K	31.21	0.967	0.095	3489K	33.42	0.967	0.103	616K
20k	30.17	0.968	0.097	985K	31.68	0.971	0.085	4467K	34.59	0.973	0.087	664K
30k	30.36	0.969	0.095	985K	31.84	0.972	0.083	4535K	34.94	0.974	0.084	664K

Iter	FisheyeGS				FisheyeGS w/ Low Thres.				3DGEER			
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Gau $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Gau $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Gau $\downarrow$
4k	23.19	0.911	0.164	677K	23.72	0.916	0.151	2778K	27.84	0.915	0.167	480K
7k	23.33	0.916	0.157	867K	24.06	0.923	0.138	3936K	28.37	0.922	0.152	621K
10k	23.45	0.918	0.153	983K	24.31	0.925	0.133	4481K	28.54	0.925	0.145	668K
20k	23.51	0.921	0.149	1072K	27.07	0.932	0.127	5272K	29.49	0.933	0.128	732K
30k	25.75	0.927	0.145	1001K	27.50	0.935	0.125	5272K	29.78	0.935	0.125	732K

Iter	FisheyeGS				FisheyeGS w/ Low Thres.				3DGEER			
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Gau $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Gau $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Gau $\downarrow$
4k	27.58	0.926	0.221	314K	28.12	0.929	0.211	1191K	28.11	0.918	0.239	292K
7k	28.47	0.935	0.203	485K	28.99	0.937	0.193	1997K	29.11	0.928	0.215	410K
10k	28.82	0.938	0.194	588K	29.29	0.941	0.181	2491K	29.73	0.935	0.198	476K
20k	29.60	0.944	0.179	760K	29.93	0.946	0.164	3563K	30.91	0.945	0.172	552K
30k	29.72	0.944	0.177	760K	29.97	0.945	0.162	3563K	31.17	0.946	0.168	552K

Iter	FisheyeGS				FisheyeGS w/ Low Thres.				3DGEER			
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Gau $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Gau $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Gau $\downarrow$
4k	23.27	0.940	0.152	602K	24.32	0.945	0.142	2429K	25.59	0.939	0.155	406K
7k	23.94	0.947	0.141	740K	25.36	0.953	0.131	3258K	27.27	0.952	0.133	524K
10k	24.12	0.950	0.137	788K	25.65	0.956	0.126	3569K	28.14	0.958	0.124	562K
20k	26.76	0.959	0.130	838K	27.77	0.961	0.120	4035K	30.39	0.967	0.108	582K
30k	26.91	0.959	0.129	838K	27.97	0.961	0.120	4035K	32.43	0.968	0.114	582K

Table K.4: **Runtime Analysis vs. Other Methods**. 3DGEER is compared with efficient splatting-based methods on MipNeRF360 and ScanNet++, with a thorough comparison in stage-wise and total runtime.

Dataset	Method	Avg. Timings (ms) $\downarrow$				
		Prep.	Dup.	Sort	Render	Total
MipNeRF360	3DGS	0.32	0.27	0.65	1.40	2.92
	3DGEER (Ours)	0.37	0.17	0.27	2.10	3.06
ScanNet++	FisheyeGS	0.10	0.63	1.45	2.33	4.70
	3DGEER (Ours)	0.13	0.26	0.59	2.89	3.98

Table K.5: **Runtime Analysis: Asso. Method on ScanNet++**. The efficiency of the PBF-based association is compared to the splatting-based EWA and UT with acceleration option SnugBox integrated into 3DGEER pipeline.

3DGEER Asso.	Prep.	Avg. Timings (ms) ↓				Total	Avg. FPS ↑
		Dup.	Sort	Render			
EWA	0.13	1.17	3.27	12.69	17.84	56	
+ SnugBox	0.13	0.60	1.42	6.22	8.68	115	
UT	0.13	0.71	1.94	7.97	11.16	90	
+ SnugBox	0.13	0.28	0.64	3.16	4.38	228	
PBF (Ours)	0.13	0.26	0.59	2.89	3.98	251	

Table K.6: **Impact of PBF-CSF Association**. The evaluation employs the artifact-sensitive LPIPS metric. To assess cross-model robustness, each variant is also tested using models trained with the other association strategies. Additionally, we report the memory cost during rendering along with the average number of Gaussian association per Tile, and the performance gap Δ LPIPS between in-method and cross-method evaluations to quantify consistency and generalization.

Asso. Method	Gaus. ↓ per Tile		Mem. ↓ (GB)	LPIPS ↓	Cross-LPIPS ↓	Δ LPIPS ↓ (1e-2)
	mean	std				
EWA	2203	1447	2.2	0.1250	0.1321	0.71
UT	1377	1058	1.4	0.1249	0.1302	0.53
PBF (Ours)	475	427	0.63	0.1245	0.1251	0.06

Table K.7: **Training-Time Transmittance Func. Replacement**. The overall results are shown in Tab. 5.

Scene	3DGS Splats + PBF				FisheyeGS Splats + PBF				3DGEER (Ours)			
	PSNR	SSIM	LPIPS	Gaus. Num (k)	PSNR	SSIM	LPIPS	Gaus. Num (k)	PSNR	SSIM	LPIPS	Gaus. Num (k)
Storage	19.27	0.859	0.173	1293	24.75	0.933	0.150	781	29.18	0.940	0.141	429
Office Room	20.89	0.848	0.182	1527	25.89	0.928	0.150	1119	29.78	0.935	0.125	732
Bedroom	27.39	0.923	0.119	1542	31.96	0.970	0.096	1079	34.94	0.974	0.084	664
Kitchen	25.15	0.890	0.239	1241	29.87	0.949	0.182	801	31.17	0.946	0.168	552
Laboratory	21.60	0.869	0.173	1379	27.03	0.962	0.131	823	32.43	0.968	0.114	582

Table K.8: **Impact of Different Supervision Space**. The overall results are shown in Tab. 6.

Supervision Space Scene	3DGEER w/o BEAP Central Perspective			3DGEER w/o BEAP Perspective			3DGEER w/o BEAP Equidistant			3DGEER (Ours) BEAP		
	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS
Storage	28.42	0.931	0.143	19.74	0.834	0.302	29.03	0.938	0.142	29.18	0.940	0.141
Office Room	28.90	0.927	0.133	21.12	0.825	0.291	29.71	0.932	0.126	29.78	0.935	0.125
Bedroom	33.86	0.965	0.090	24.38	0.882	0.240	34.60	0.973	0.096	34.94	0.974	0.084
Kitchen	30.65	0.941	0.172	23.23	0.889	0.338	31.23	0.941	0.177	31.17	0.946	0.168
Laboratory	27.35	0.951	0.118	17.08	0.834	0.329	30.65	0.956	0.135	32.43	0.968	0.114

Table K.9: **ZipNeRF Quantitative Results by Sequence.** Alameda, Berlin, London, and NYC in order. The overall results are shown in Tab. 3

Train Data	1/8 FE		1/8 FE		1/8 FE		1/4 PH		1/4 PH		1/4 PH	
Test Data	1/8 FE	1/4 PH	1/8 FE	1/4 PH	1/8 FE	1/4 PH	1/8 FE	1/4 PH	1/8 FE	1/4 PH	1/8 FE	1/4 PH
Method	PSNR $\uparrow$		SSIM $\uparrow$		LPIPS $\downarrow$		PSNR $\uparrow$		SSIM $\uparrow$		LPIPS $\downarrow$	
FisheyeGS	20.74	23.64	0.820	0.802	0.223	0.277	17.39	23.74	0.736	0.839	0.265	0.240
EVER	22.46	23.06	0.852	0.777	0.144	0.274	20.06	22.83	0.789	0.794	0.213	0.250
3DGUT	21.78	22.39	0.841	0.718	0.193	0.368	17.03	22.58	0.588	0.765	0.329	0.311
Ours	24.60	25.00	0.857	0.839	0.154	0.241	21.49	25.21	0.792	0.807	0.232	0.277

Train Data	1/8 FE		1/8 FE		1/8 FE		1/4 PH		1/4 PH		1/4 PH	
Test Data	1/8 FE	1/4 PH	1/8 FE	1/4 PH	1/8 FE	1/4 PH	1/8 FE	1/4 PH	1/8 FE	1/4 PH	1/8 FE	1/4 PH
Method	PSNR $\uparrow$		SSIM $\uparrow$		LPIPS $\downarrow$		PSNR $\uparrow$		SSIM $\uparrow$		LPIPS $\downarrow$	
FisheyeGS	24.25	28.25	0.895	0.924	0.206	0.209	20.08	27.86	0.848	0.929	0.233	0.194
EVER	25.26	26.97	0.912	0.907	0.155	0.221	23.18	27.72	0.877	0.913	0.196	0.212
3DGUT	25.37	27.02	0.904	0.876	0.197	0.288	18.57	27.43	0.730	0.898	0.297	0.249
Ours	25.51	29.15	0.916	0.925	0.166	0.194	23.58	29.51	0.878	0.915	0.219	0.228

Train Data	1/8 FE		1/8 FE		1/8 FE		1/4 PH		1/4 PH		1/4 PH	
Test Data	1/8 FE	1/4 PH	1/8 FE	1/4 PH	1/8 FE	1/4 PH	1/8 FE	1/4 PH	1/8 FE	1/4 PH	1/8 FE	1/4 PH
Method	PSNR $\uparrow$		SSIM $\uparrow$		LPIPS $\downarrow$		PSNR $\uparrow$		SSIM $\uparrow$		LPIPS $\downarrow$	
FisheyeGS	23.57	26.65	0.829	0.861	0.253	0.254	20.83	27.33	0.768	0.883	0.288	0.229
EVER	24.59	26.83	0.823	0.859	0.218	0.223	23.14	26.98	0.793	0.860	0.268	0.235
3DGUT	25.61	26.25	0.854	0.803	0.221	0.336	19.82	26.41	0.669	0.833	0.313	0.291
Ours	26.60	27.84	0.895	0.882	0.142	0.230	23.95	27.15	0.840	0.846	0.217	0.289

Train Data	1/8 FE		1/8 FE		1/8 FE		1/4 PH		1/4 PH		1/4 PH	
Test Data	1/8 FE	1/4 PH	1/8 FE	1/4 PH	1/8 FE	1/4 PH	1/8 FE	1/4 PH	1/8 FE	1/4 PH	1/8 FE	1/4 PH
Method	PSNR $\uparrow$		SSIM $\uparrow$		LPIPS $\downarrow$		PSNR $\uparrow$		SSIM $\uparrow$		LPIPS $\downarrow$	
FisheyeGS	24.15	27.21	0.890	0.885	0.162	0.215	19.41	27.50	0.811	0.907	0.203	0.183
EVER	28.39	27.71	0.931	0.860	0.095	0.228	25.22	28.27	0.879	0.883	0.154	0.192
3DGUT	26.31	26.69	0.917	0.817	0.121	0.302	19.00	27.42	0.659	0.866	0.263	0.228
Ours	28.26	28.49	0.922	0.906	0.097	0.190	24.52	28.55	0.874	0.885	0.168	0.220